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Coupling Authorization and Payment Edges in One PageRank Walk for Ranking Smart Contracts: Commit-Gated Rules and a Contract-Graph Re-Analysis on an Ethereum Rollup

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ABSTRACT Wallets, explorers and integrators must judge which of a rollup's million-plus smart contracts users rely on. Ledger PageRank variants rank addresses by payments, walk over user accounts that cost nothing to create, and omit token allowances, by which owners authorize spenders; we found no pre-specified test of a walk over both relations. We test on Arbitrum One whether a random walk mixing authorization and payment edges at every node ranks contracts better than a walk on either alone. Two decision rules were committed to the authors' Git repository before this study extracted their evaluation logs, the second eight seconds before a commit-checking script queried its window. The main analysis, a re-analysis defined after the registered results were known, uses contract-only graphs, US-dollar amounts on five tokens, one-year score windows and allowances capped at the owner's balance. Neither rule was met in either analysis. On 7,915 contracts the coupled walk ranked later approvals better than the payment walk (rank-correlation difference 0.074, 95% interval 0.060 to 0.087) but later payers worse than the 0.01 margin allows (−0.016, interval −0.022 to −0.012). Post hoc, counts of approving and of paying contracts beat the coupled walk on later approvals and payers, and about 374 farm contracts holding a dollar each would reach the cohort's top authorization score. We conclude that coupling did not improve on simpler scores; wallets, explorers and integrations can show the two counts as weak hints and should leave contracts without contract-to-contract edges, about 27% here, unranked.

INDEX TERMS Blockchain, decentralized finance, PageRank, reputation systems, smart contracts, Sybil attacks, token allowances.

I. INTRODUCTION

AN Ethereum rollup holds a very large number of smart contracts. In eighteen months of events of five widely used tokens on Arbitrum One [1], 1,829,719 addresses hold contract code and are not user accounts (Section V-B). A wallet that shows a user what a transaction will touch, an explorer that lists contracts and a developer who chooses contracts to integrate all need to know which of these contracts users rely on. The ledger records two relations that bear on the question. A contract is paid when tokens are transferred to it, and it is authorized when an owner lets it spend the owner's tokens: under the ERC-20 (Ethereum Request for Comments 20) token interface an owner calls `approve` to let a spender draw up to a stated amount, and the `permit` function of Ethereum Improvement Proposal (EIP) 2612 sets the same allowance from an off-chain signature [2], [3]. An allowance moves no funds; it grants a capability that the spender may exercise later, and many allowances are unlimited [4].

Ranking by links has a long record on the web. PageRank orders pages by the stationary distribution of a random walk over hyperlinks [5], and Cho, Garcia-Molina and Page showed that the order in which a crawler visits pages decides how soon it reaches the important ones [6]. On ledgers the idea has been applied almost entirely to payments and to all

addresses at once. Do and Do read PageRank and HodgeRank on Ethereum transactions as a measure of social credit [7]; Adaptive Weighted PageRank (AWP) of Do, Do and Nguyen weights every transfer by its value and age and restarts the walk at recently active senders [8]; RiskProp propagates account risk through the transaction network [9]. Surveys of blockchain reputation systems find the same reliance on payment volume or on explicit ratings [10]. Three things are missing for the question above. These walks leave out allowances, although allowances link the same addresses. Their graphs hold user accounts, which cost no deployment and whose controllers such a study does not measure, and user accounts hold most of the allowances toward a contract (Section V-F), so a walk over all accounts is largely a walk over them. And we found no test of a walk over both relations under rules that the study fixed before it extracted its evaluation data.

We ask whether one random walk over both relations ranks contracts better than a walk over either relation alone, and in particular whether adding authorization edges improves on a walk over payments. Payments and authorizations form two layers over one set of nodes, the setting of multilayer ranking [11], [12]. We define a coupled PageRank (C-PR) that blends the two transition rows of each node with a fixed weight λ on the authorization layer, and an ablation, endorsement-seeded PageRank (S-PR), that walks the payment layer but draws its restarts from the authorization walk, as TrustRank draws its restarts from a vetted seed set [13]. The end points of the C-PR family, $\lambda = 1$ and $\lambda = 0$, walk one layer alone and are the comparators of both decision rules. In the main analysis every node of every graph is a contract: user accounts, smart accounts among them, are nodes of no graph, restarts run over contracts, and an allowance counts only up to the balance its owner holds.

The evaluation looks forward in time. Scores are frozen at a date and judged by their Kendall rank correlation [14] with what arrives afterwards: the new owners that approve a contract and the new addresses that pay it. Both labels are arrivals of the same kinds of edge that build the layers, and both count every account, user accounts included, so they measure later use by accounts that no score reads.

Two decision rules were fixed before this study extracted the data that judge them. The analysis plan, which fixes Rule A and every score, was committed to the authors' Git repository ten seconds before the BigQuery job that copied the observation window was created. The registration of the second window, which fixes Rule B with a non-inferiority margin, was committed eight seconds before the job that copied the logs of July to September 2026, and the script that runs that job refuses to start unless the registration files are committed without local edits; it then records their SHA-256 hashes in a manifest. We call this arrangement commit-gated. The precedence of both commits over the extractions rests on commit and job times, not on a third-party timestamp (Section IV-C). The plan and the registration fixed an analysis in raw token units over every ERC-20 token, on graphs that hold every account of the cohort's events; we call it the registered analysis. The main analysis of this paper, the contract-graph analysis, was defined after the registered results were known (Section IV-D). Its verdicts are therefore a re-analysis under the same rules, none of them is a registered result, and the registered analysis is reported beside it in the tables and figures of the rules. This paper applies Rule A and two of the registered criteria of Rule B, F1 and F2; under the registration the rule holds only if each of its criteria holds. Neither rule is met in either analysis.

The contributions are the following.

1) A two-layer model for ranking contracts with a coupled walk, a seeded ablation and two analytic remarks: for λ strictly between 0 and 1 the weight acts only through nodes with out-edges in both layers, and for a spender approved by owners that nobody approves the authorization walk reduces to an owner-normalized in-strength.
2) A commit-gated protocol for ranking experiments on ledger data, with the record that dates each commitment (Section IV) and the limits of that record: commit times are set by the committer's clock, and no third-party timestamp anchors them.
3) A negative result on Arbitrum One, on graphs whose nodes are contracts only. C-PR met neither rule, in the contract-graph analysis or in the registered one. On the contract graph it ranked later approvals below the authorization walk and later payers below the payment walk in the first window; in the second window its gain over the payment walk on later approvals came with a loss on later payers beyond the registered margin. In contrasts computed after the labels were known, C-PR ranked later approvals below the count of approving contracts and later payers below the count of paying contracts in both windows; against the union in-degree, which counts the edges the coupled walk follows, it lay slightly above on later approvals and below on later payers.
4) A calibration of known link-farm results to an authorization layer whose nodes are contracts. A farm address must be deployed and must hold a balance for its approval to count. On the contract graph a closed farm collects 6.16 times its population share at damping 0.85, about 374 farm contracts reach the highest authorization score of the cohort, and a balance threshold thins the honest graph faster than it burdens a farm.

A companion manuscript [15] reports the discovery-order test, a crawl-ordering experiment on the same contract graph that asks whether ordering a crawl by a walk brings it to the contracts with the most user-account demand earlier than ordering it by a count, and it sets the single-layer walks, EndorseRank and AWP, against the counts they smooth. Both papers describe the same data, account types and graphs, and both report the spender holdout coefficients of the raw counts, EndorseRank and AWP with their count contrasts, the closed

form of the authorization walk and the solver runtimes, with the same values. The discovery-order test is reported only there; Section VI-H quotes what it found about C-PR. C-PR's two rules, the weight sweep, the seeded ablation and the Sybil calibration are reported only here. Section II reviews related work, Section III defines the model, Section IV states the rules and their record, Section V describes data and measurement, Section VI reports the results, Section VII calibrates farm attacks, Section VIII discusses threats to validity, Section IX draws implications for wallets, explorers and integrations, and Section X concludes. Appendix A collects exploratory checks.

II. RELATED WORK

A. REPUTATION FROM LEDGER GRAPHS

PageRank ranks the nodes of a directed graph by the stationary distribution of a damped random walk [5], [16]. Hyperlink-Induced Topic Search (HITS) separates hubs from authorities [17], and EigenTrust normalizes local trust values so that peers in a file-sharing network cannot inflate their own standing [18]. On public ledgers these ideas have been applied almost entirely to transfers. Besides the methods named above [7], [8], network embeddings of transaction graphs detect phishing accounts better than per-account features [19], and a survey of knowledge discovery in cryptocurrency transactions lists degree- and value-based centralities as the usual address descriptors [20]. Approvals appear in the literature in two roles: as a clustering heuristic that links addresses of one entity [21], and as a security risk when unlimited [4]. Relations among contracts have been studied as compositions of decentralized-finance protocols [22], not as a ranking. Deployed tooling such as Human Passport scores the holder of an address from weighted credentials and does not rank relations between addresses [23]. The companion study [15] uses the latest allowance as the edge of a ranking and repeats the crawl-ordering experiment of Cho, Garcia-Molina and Page [6] on the contract graph; the present paper combines that edge with the payment edge in one walk.

B. MULTILAYER AND PERSONALIZED RANKING

Multiplex PageRank lets the centrality a node earns in one layer bias the walk in another [11], and the interconnected-multilayer framework of De Domenico *et al.* lets a walker move inside layers and switch between them, ranking nodes by their versatility across layers [12]. Topic-sensitive and personalized PageRank change the teleportation vector rather than the transition matrix [24], [25], and TrustRank confines teleportation to a hand-checked seed set to demote web spam [13]. C-PR is a special case of the first family: a per-node convex combination of two row-stochastic matrices with one global weight and no copies of a node across layers. S-PR belongs to the second family. Proposition 1 below is a first-order perturbation bound of the standard kind for PageRank [16], specialized to this convex combination. We have not found in the literature a test of either construction on two on-chain relations among the same addresses, judged by a rule that the study fixed before it extracted its evaluation data.

C. DEGREE BASELINES

Under preferential attachment new links go disproportionately to nodes that already have many [26], and measurements of growing networks show the same tendency [27]. Degree-based scores are therefore standard baselines in link prediction [28], including on Ethereum transaction networks [29], and on graphs with weak degree correlations PageRank is on average proportional to in-degree [30]. A claim that a walk adds information should thus be tested against the raw count it smooths. We set every walk against that count, and the coupled walk also against the union in-degree, which counts the in-edges of both layers together.

D. MANIPULATION AND SYBIL RESISTANCE

Where identities are free, one actor can appear as many [31]. Gyöngyi and Garcia-Molina derive the link farms and alliances that raise a target's PageRank the most [32]. Cheng and Friedman show that no symmetric reputation function, PageRank included, is Sybil-proof, whereas some asymmetric functions computed from flows or paths out of a trusted source are [33]. Section VII applies these results to the authorization layer; it calibrates them rather than deriving new ones. On ledgers, wash trading inflates payment activity [34], [35], studies of airdrops measure and detect the addresses that hunters create in number [36], [37], proof-of-personhood schemes trade Sybil resistance against privacy [38], and a recent review of decentralized systems names credential farming among the remaining weaknesses of such gates [39].

E. PRE-SPECIFIED ANALYSES

Preregistration separates the analyses that test a hypothesis from those that generate one, and it can be applied to data that already exist, provided the analysts have not yet examined the outcomes [40]. With observational ledger data the events of a test window exist on chain from the moment they occur, so the strongest practical guarantee a study can give is that it had not extracted them when its rule was fixed. Construct-validity theory adds a caution that matters below: a score should agree with checks of what it was designed to track, so such agreement is not independent evidence [41].

III. MODEL

A. TWO LAYERS ON ONE SET OF CONTRACTS

Let $V = V_A \cup V_T$ be the set of contracts in either layer (Fig. 1). Every node is a contract, an address with code that is not a user account under the rule of Section V-B. Externally owned accounts, accounts with an EIP-7702 delegation and smart accounts are user accounts: they are nodes of no graph, and an approval or a transfer with a user account at either end is an edge of no graph. We use *authorization* for the layer, its edges and its walk, *approval* for the event that creates an edge, and *allowance* for the approved amount; likewise *payment* for the second layer and *transfer* for its events.

An authorization edge $u \to v$ exists when, for at least one token, the latest `Approval` event from owner contract u to spender contract v in the score window carries a positive value and u holds a positive balance of that token at the freeze. The edge counts the allowance only up to that balance, since a spender cannot draw more than the owner holds: the capped allowance is $a^{\text{eff}} = \min(a^\star, \text{bal}_u)$, in US dollars at the price of the freeze day, with $a^\star$ the latest allowance. We call this the balance cap. A latest value of zero deletes the pair, and an allowance whose owner holds no balance is capped to zero and is no edge. This is what the pipeline measures, and it differs from the allowance in force in three ways: allowances set before the start of the score window and not re-emitted inside it are invisible; a finite allowance that `transferFrom` has since drawn down keeps its approved value, because a token need not emit an `Approval` event when an allowance is spent; and an owner who approves an allowance manager such as Uniswap's Permit2 appears with an edge to that manager only. A payment edge $u \to v$ exists when contract u sent contract v at least one transfer with a positive amount in the score window. Native ether moves without `Transfer` logs and enters the payment layer only as wrapped ether.

The operator studied here weights both layers by amounts:

$$\tilde{w}_A(u,v) = \sum_{\text{token}} \min\big(a^{\text{eff}}(u,v,\text{token}),\ U\big), \tag{1}$$

$$\tilde{w}_T(u,v) = \sum_{e:\,u\to v} x_e\,\sigma(\Delta t_e), \tag{2}$$

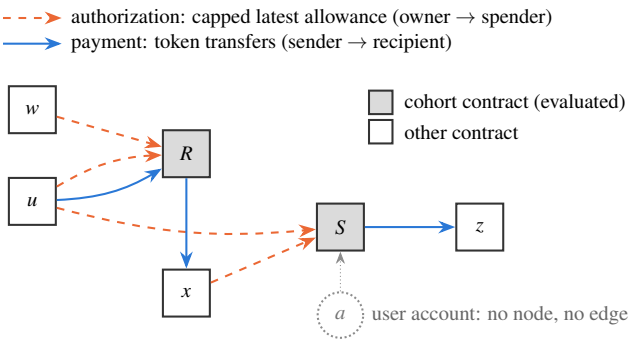
where x_e is the amount of transfer e in US dollars at the price of its day and $\sigma(\Delta t) = 1/(1+\exp(k(\Delta t - t_0)))$ a logistic discount on the age Δt in days, the form used by AWP [8]; that paper fixes no parameter values, and $k = 0.01$ and $t_0 = 180$ are our choices. Ages are measured back from the freeze. The authorization layer is not discounted. The bound $U = 16{,}000$ dollars limits what one allowance can add to a row (Section V-D). The plan fixed raw token base units, every ERC-20 token, no cap and graphs that hold every account of the cohort's events; that is the registered analysis. Write $o_A(u)$ and $o_T(u)$ for the out-strengths and P_A, P_T for the row-normalized matrices, whose rows are zero where a node has no out-edge in that layer.

B. PAGERANK WITH A RESTART DISTRIBUTION

Every score below is the fixed point of

$$\mathbf{r}^{(t+1)} = (1-d)\,\boldsymbol{\pi} + d\,P^\top \mathbf{r}^{(t)} + d\,\big(\mathbf{1}_{\text{dang}}^\top \mathbf{r}^{(t)}\big)\,\boldsymbol{\pi}, \tag{3}$$

where P is the transition matrix, $\boldsymbol{\pi}$ the restart distribution over the contracts of the graph, and $\mathbf{1}_{\text{dang}}$ the indicator of nodes with an all-zero row, whose mass is sent back through $\boldsymbol{\pi}$. We use $d = 0.85$, start from the uniform vector, and stop when the ℓ_1 change falls below 10^{-8} or after 300 iterations.



$u \in B$: C-PR row $= \lambda P_A(u,\cdot) + (1-\lambda) P_T(u,\cdot)$
w, x: authorization out-edges only, follow P_A for $\lambda > 0$
R, S: payment out-edges only, follow P_T for $\lambda < 1$

FIGURE 1. The two layers on one set of contracts (schematic). Every node is a contract; the shaded squares are cohort contracts, the ones evaluated. A user account, such as *a*, is no node, and its approvals and transfers are edges of no graph, although the labels count them. Only *u* has out-edges in both layers, so only its row depends on λ for $0 < \lambda < 1$. At $\lambda = 1$ the walk reads no payment edge and *z* is no part of its graph, so *R* and *S* are dangling; at $\lambda = 0$ it reads no authorization edge, so *w* is no part of its graph and *x* is dangling. S-PR walks the solid edges only and restarts each jump in proportion to the authorization walk's score.

C. COUPLED PAGERANK

C-PR mixes the two rows of each node, renormalizing over the layers in which the node has out-edges:

$$P^{(\lambda)}_{u\cdot} = \alpha_A(u)\,P_{A,u\cdot} + \alpha_T(u)\,P_{T,u\cdot}, \tag{4}$$

$$\alpha_A(u) = \frac{\lambda\,\mathbf{1}[o_A(u)>0]}{D_\lambda(u)}, \quad \alpha_T(u) = \frac{(1-\lambda)\,\mathbf{1}[o_T(u)>0]}{D_\lambda(u)}, \tag{5}$$

with $D_\lambda(u) = \lambda\,\mathbf{1}[o_A(u)>0] + (1-\lambda)\,\mathbf{1}[o_T(u)>0]$. A node with out-edges in one layer follows that layer with weight one, and a node with none in either is dangling. Restarts are uniform over the contracts. The primary setting $\lambda = 0.5$ and the sensitivity values 0.25 and 0.75 were fixed in the plan (Section IV).

At $\lambda = 1$ the walk reads the authorization layer alone, and the contracts that only the payment layer contains are no part of its graph; at $\lambda = 0$ it reads the payment layer alone. We call these the authorization walk and the payment walk; they are the comparators of both rules. The family is discontinuous at both ends. For every λ strictly between 0 and 1, a node whose out-edges are all payments, such as a contract that pays out but approves no one, follows the payment layer, whereas at $\lambda = 1$ the same node is dangling or absent. Symmetrically, a node whose out-edges are all authorizations follows the authorization layer at every interior λ but is dangling or absent at $\lambda = 0$. Unit tests on small fixtures check that each end point reproduces its single-layer walk.

D. SEEDED ABLATION

S-PR keeps the payment walk and draws its restarts from the authorization walk's score $\mathbf{r}_A$, restricted to V_T and renormalized:

$$\pi_u = \frac{r_{A,u}}{\sum_{v\in V_T} r_{A,v}}, \qquad u \in V_T, \tag{6}$$

with $r_{A,u} = 0$ for $u \notin V_A$ and a uniform fallback if the denominator vanishes. Authorization then decides where the walker lands after a jump, and payments decide where it goes between jumps. Comparing S-PR with C-PR separates the authorization edge as a route from the authorization edge as a seed.

E. REFERENCE SCORES

Two walks and four counts serve as references. AWP [8] walks the payment layer with weights $\sigma(\Delta t_e)V(x_e)$ summed over transfers, where $V(z) = 2/(1+e^{-bz}) - 1$ is a bounded value transform, and restarts at each contract in proportion to its recent sending activity. EndorseRank [15] walks the authorization layer with the same bounded weights, applied to the capped allowance and the age of its approval, and with uniform restarts; a variant restarts in proportion to approving activity. In the main analysis $b = \ln 3/m$, so that an amount of m weighs one half, with $m_T = 160$ dollars for transfers and $m_A = 0.071$ dollars for capped allowances (Section V-D); in the plan $b = 1$ on raw base units, which makes $V(z)$ within 10^{-4} of one for any amount of ten base units or more. Both reference walks bound the weight of each event, whereas C-PR lets amounts matter up to U. The counts are taken on the contract graph at the freeze: the in-approve degree, the number of owner contracts that hold a capped allowance to the contract; the in-approve value, the sum of $\tilde{w}_A$ over those owners; the transfer in-degree, the number of other contracts that sent it a transfer; and the union in-degree, the number of other contracts with an edge to it in either layer, which is the count of the edges that C-PR walks. No count reads a user account.

F. WHERE THE WEIGHT ACTS

Let $B = \{u : o_A(u) > 0 \text{ and } o_T(u) > 0\}$ be the set of nodes with out-edges in both layers.

Proposition 1: For $\lambda \in (0,1)$ and uniform $\boldsymbol{\pi}$, the rows of $P^{(\lambda)}$ outside B do not depend on λ, and the derivative $\mathbf{r}'(\lambda)$ of the stationary vector of (3) with respect to λ satisfies

$$\mathbf{r}'(\lambda) = d\left(I - d\,G_\lambda^\top\right)^{-1} \sum_{u \in B} r_u \left(\mathbf{p}_{A,u} - \mathbf{p}_{T,u}\right), \qquad (7)$$

where $\mathbf{p}_{A,u}$ and $\mathbf{p}_{T,u}$ are the transposed rows of P_A and P_T and G_λ is $P^{(\lambda)}$ with each dangling row set to $\boldsymbol{\pi}^\top$. Hence $\|\mathbf{r}'(\lambda)\|_1 \le 2d\, r(B)/(1-d)$ with $r(B) = \sum_{u \in B} r_u$.

Proof: Outside B one indicator in (5) is zero, so the row is a single layer's row or zero whatever λ is, and the dangling set does not change on $(0,1)$. For $u \in B$, $D_\lambda(u) = 1$ and the row is $\lambda P_{A,u\cdot} + (1-\lambda)P_{T,u\cdot}$, whose derivative is $P_{A,u\cdot} - P_{T,u\cdot}$. Because $\mathbf{r}$ sums to one, (3) can be written $\mathbf{r} = (1-d)\boldsymbol{\pi} + d\,G_\lambda^\top \mathbf{r}$; differentiating gives (7). The matrix $G_\lambda^\top$ is column-stochastic, so $\|(I - dG_\lambda^\top)^{-1}\|_1 \le 1/(1-d)$, and each difference of two probability rows has ℓ_1 norm at most 2. ∎

The proposition says where the weight can act, not that its action is small: when $r(B)$ is large the bound exceeds the trivial bound of 2 on the ℓ_1 distance between two distributions. In the contract graph few nodes can belong to B, because only an owner contract has an authorization out-edge: the authorization layer has 9,459 owner contracts at the freeze of 30 June 2026 and 10,153 at that of 31 March 2026, in union graphs of 1,299,438 and 1,111,554 contracts (Table 4). We did not compute $r(B)$; Appendix A reports how far the rankings move between interior values of λ.

G. THE AUTHORIZATION WALK ON DEPTH-ONE SPENDERS

In the authorization layer between contracts most nodes either only grant approvals or only receive them: at the freeze of 30 June 2026 it has 9,459 owner contracts and 100,112 spender contracts on about 109,000 nodes, so few contracts are both. The companion study derives a closed form for this structure [15]; we restate it with its proof.

Proposition 2: Under (3) with uniform $\boldsymbol{\pi}$, every node without in-edges has the same score c, and a node v whose in-neighbors all lack in-edges scores $r_v = c\,(1 + d\,\delta(v))$, where $\delta(v) = \sum_u w(u,v)/o(u)$ is its owner-normalized in-strength.

Proof: With uniform restarts and dangling mass r_D, (3) reads $r_v = (1-d+d\,r_D)/|V| + d\sum_u r_u\, w(u,v)/o(u)$. A node without in-edges receives only the first term, which we call c. If every in-neighbor u of v lacks in-edges, then $r_u = c$, and substituting gives $r_v = c + d\,c\,\delta(v)$. ∎

The statement holds for any edge weights, so it covers EndorseRank and C-PR at $\lambda = 1$, each with its own δ. On the EndorseRank graph at the freeze of 30 June 2026, 926 contracts of the matched cohort have an authorization in-edge. Of these, 651 have depth one, and the computed score lies within 1% of the closed form for every one of them; over all 926 the share is 80.3%, and 33.8% of the deeper ones [15]. In the registered analysis the figures are 20,942 of 26,512, every one of them, 86.4% and 35.1%. For most of these spenders the authorization walk is therefore a reweighted count of approving owners, and C-PR inherits that limit wherever it follows the authorization layer.

H. COST

One iteration of (3) costs $O(|E|)$ [42], and C-PR walks the union of both edge sets. Table 1 reports the solver time of each score at the freeze of 30 June 2026, with $|V|$ and $|E|$ of each solver graph. In the main analysis the timed graph is the whole contract graph, that is, all pairs between contracts whether or not they touch the matched cohort; in the registered analysis it is the graph of the cohort's events. In the main analysis C-PR takes 3.08 s on 1,299,479 nodes and 2,591,253 edges, 1.09 times AWP's time on 1.004 times its edges; in the registered analysis 105.51 s on 25,304,106 nodes and 71,375,442 edges, 1.51 times AWP's time on 1.19 times its edges. EndorseRank is the fastest score because its graph is the sparsest: the authorization graph holds 4.5% of the payment graph's edges in the main analysis and 37.4% in the registered one. The time ratio of AWP to EndorseRank, 27.9 and 2.36, combines that edge ratio (22.4 and 2.68) with the iteration counts of Table 1: AWP needs 98 iterations to EndorseRank's 56 in the main

TABLE 1. Solver Time at the Freeze of 30 June 2026 on the Whole Contract Graph and on the Graphs of the Registered Analysis (Mean and SD of Five Timed Runs After One Warm-Up)

Score	$\|V\|$	$\|E\|$	Iter.	Time (s)
Main analysis: the whole contract graph				
EndorseRank	109,381	114,997	56	0.101 (0.009)
AWP	1,298,649	2,580,381	98	2.821 (0.189)
C-PR ($\lambda = 0.5$)	1,299,479	2,591,253	92	3.080 (0.184)
S-PR	1,298,724	2,580,709	153	2.657 (0.202)
Registered analysis: the graphs of its cohort's events				
EndorseRank	13,055,150	22,321,470	84	29.60 (0.61)
AWP	23,029,298	59,737,617	93	69.82 (2.30)
C-PR ($\lambda = 0.5$)	25,304,106	71,375,442	92	105.51 (2.48)
S-PR	23,029,298	59,737,617	185	100.86 (4.82)

One session of one warm-up and five timed runs per graph and setting, the measurement cited throughout; the time covers matrix assembly and power iteration ($d = 0.85$, tolerance 10^{-8}, at most 300 iterations) on edge lists built in advance. S-PR includes the solve of the authorization layer that supplies its restart vector. $|V|$ and $|E|$ are those of each solver graph, which holds the pairs of positive weight (Section V-E); C-PR walks the union of both layers. In the main analysis the timed graph is the whole contract graph (all pairs between contracts, whether or not they touch the matched cohort); the registered graphs hold the events that name a contract of the registered cohort, user accounts included. Same machine for both analyses (Section III-H).

analysis and 93 to 84 in the registered one. The speed reflects the smaller edge count and the iterations that graph needs, not the algorithm, and any walk that uses the payment layer pays for its edges. Times are means of five runs after one untimed warm-up on one machine (Intel Core Ultra 9 185H, 64 GB of memory, Windows 11, Python 3.11.9, NumPy 2.3.2, SciPy 1.16.1).

IV. PRE-SPECIFIED RULES AND THEIR RECORD

Table 2 lists the commitments in the order in which they were made, the record of each one, and what was known at each point. Three features of this history matter for interpretation. First, Rule A and the coupled operator were fixed before any log of this study was extracted, so no score or label of this study had been computed. Second, the comparison that motivated Rule B was noticed in the first window after its labels were known, so the evidence from that window is exploratory. Third, the main analysis was defined after every result of the registered analysis was known; it applies the same rules and thresholds, but its verdicts are not registered tests. Section VI reports it first, with the registered analysis beside it.

A. RULE A (ANALYSIS PLAN, 9 OCTOBER 2026, 13:41:27 UTC)

Rule A asked whether C-PR beats both layers walked alone. Its primary part used the first holdout window, W0 (scores frozen on 31 March 2026, labels from April to June 2026): at $\lambda = 0.5$, C-PR should do no worse than the authorization walk on new approval pairs, meaning that the 95% interval of $\Delta\tau = \tau_{\text{C-PR}} - \tau_{\lambda=1}$ contains zero or lies above it (A1), and should beat the payment walk on new transfer senders, meaning that the interval of $\tau_{\text{C-PR}} - \tau_{\lambda=0}$ lies above zero (A2). Its secondary part used the matched cohort in the window of the scores themselves: C-PR's point estimate had to lie inside the 95% interval of the better single layer on a family of transfer checks and on a family of authorization checks, and C-PR had to exceed both layers on a family of three Sybil-adjusted stability checks (the plan's name; the share of distinct inbound counterparties, tenure and active months). If the primary part failed, the question whether C-PR beats a walk over either layer alone was to be answered in the negative, and no λ was to be tuned after labels were seen.

Two criteria of the rule are weak by design. "No worse" as defined rewards imprecision, since a wider interval contains zero more easily, and the secondary part compares a point estimate with another score's marginal interval rather than using a paired contrast. Anyone reusing the protocol should set a non-inferiority margin for the first and paired contrasts for the second; we report the paired contrasts beside the rule.

B. RULE B (REGISTRATION, 9 OCTOBER 2026, 16:49:32 UTC)

Rule B asked whether C-PR improves on the payment walk, $\Delta\tau = \tau_{\text{C-PR}} - \tau_{\lambda=0}$, in a second window, W1, frozen on 30 June 2026 with labels counted from 1 July to 30 September 2026. This paper applies two of the registered criteria of Rule B, F1 and F2; under the registration the rule holds only if each of its criteria holds.

- F1 (spenders; new approval pairs), superiority: the whole 95% interval of $\Delta\tau$ must be positive;
- F2 (spenders; new transfer senders), non-inferiority: the lower end of the interval must lie above $-\delta$.

Because the rule is an intersection of its criteria, a failure of F1 or F2 settles that it is not met, and testing each criterion at the 5% level keeps the probability of a false claim of improvement at or below 5% without a multiplicity correction. Had both held, the two alone could not have shown the rule met, and we would have reported only that both applied criteria hold. The margin $\delta = 0.01$ is the half-width of the W0 interval of the contrast that F2 repeats (-0.030, $[-0.034, -0.026]$, half-width 0.0040), rounded up to two decimals. It measures the precision of a spender contrast, not a difference judged to matter in practice.

With the rule the registration fixed F6, the primary part of Rule A rerun on W1, to be reported outside the decision, and two sensitivity analyses, which we report for F1, F2 and F6: S1 takes AWP as the comparator of F1 and F2, and S2 keeps cohort contracts that a graph does not contain as isolated nodes rather than scoring them zero. Neither enters the decision; their verdicts are nominal results with unadjusted 95% intervals. Scores, solver, queries, cohort filter and bootstrap (400 paired resamples, seed 42) were those of the plan.

C. GATE ON EXTRACTION

The configuration carries a status field for W1, which the registration commit set to "registered". Before it queries anything, the extraction script checks that the registration and

TABLE 2. Order of Commitments, the Record of Each, and What Was Known When It Was Made (2026, Times in UTC)

When	Commitment or event	Record	Already known
9 Oct, 13:41:27	Analysis plan and configuration committed: cohort rule, scores (C-PR at $\lambda \in \{0.25, 0.5, 0.75\}$ and its end points, S-PR), W0 labels, statistics and rule A; amounts in raw base units, every ERC-20 token	`d060f3a`	No log of this study extracted
9 Oct, 13:41:37–13:43:54	Scan A of the observation window; the BigQuery job was created 10 s after the plan commit	Job metadata; cost ledger	—
9 Oct, before 16:49	Cohort of 27,844 contracts, extraction of the events that name one of them, same-window and W0 evaluation; rule A judged	Summaries committed in `1747e64`	Logs of the observation window
9 Oct, 16:49:32	Registration of W1: rule B with $\delta = 0.01$, of which this paper applies the criteria F1 and F2; F6 and two sensitivity analyses to be reported beside it; configuration status set to registered	`1747e64`	Every same-window and W0 result of the registered analysis
9 Oct, 16:49:33	Extraction script checks the committed registration and writes its manifest (1 s after the commit): commit and both SHA-256 hashes	Manifest	—
9 Oct, 16:49:40–16:50:11	Scan B of July–September 2026; job created 8 s after the commit; manifest closed at 16:50:55	Job metadata; manifest	No July–September log extracted for this study before this
9 Oct, 18:00:55	Registered W1 evaluation, then robustness and Sybil model, committed	`344017e`	W1 labels
9 Oct, 23:24:39	Registered benchmark and data description (BigQuery job times, amount shares) committed	`7054746`	W1 labels
Later	After the registered results were known, the revised analysis was defined: amounts in US dollars on five tokens; scores from the twelve months before each freeze; graphs whose nodes are contracts other than smart accounts, with edges read from the whole chain; each allowance counted up to its owner's balance at the freeze, with balances reconstructed from `Transfer` logs; constants set on the contract graph; and the discovery-order test. It is the main analysis of this paper. Rule A and the criteria F1 and F2 of rule B are applied to its scores as a re-analysis; the discovery-order test is reported in [15]; the exploratory checks of Appendix A followed	Definitions and results in the public repository (Table 3)	Every registered result

Commit times are committer times, recorded in UTC+9 and converted. Job times are the creation and end times that BigQuery reports for each job. Commit identifiers are those of the public repository; the full identifiers, those of the authors' repository and the file hashes are in Table 3. The last row carries no date: no definition of the main analysis preceded the registered results.

the configuration are committed without local changes and that the status is "registered"; it then records the commit and the SHA-256 hashes of both files in a manifest. The gate compares no hash; the hashes let a reader check afterwards that the committed files are the ones the script saw. The script has no override, and the manifest's field `override_used` is a constant that records nothing. Table 3 gives the full identifiers of the commits and the hashes. The authors' Git repository is private. The folder of the study is published with every commit that changed it at https://github.com/abitocodes/ERC20-Allowance-PageRank-Wallet-Reputation, where each commit of the registered record keeps the author, committer, times, message and file contents of its commit in the authors' repository but has a new identifier, because it holds only that folder. Tables 2, 3 and 7 give the public identifiers. The manifest and the other files written in the authors' repository record the identifiers of that repository (the manifest names the registration commit `0d3a057`, which is `1747e64` in the public repository), and Table 3 and the repository's `HISTORY.md` list both. The manifest hashed the files on the authors' Windows checkout, with CRLF line ends; the same text with LF line ends, as git stores it, gives the second pair of hashes, and both were checked against the committed files.

What the record shows depends on who keeps it. The commit times are set by the clock of the machine that made the commits and can be chosen by their author. The BigQuery job times (creation of scan A at 13:41:37 and of scan B at 16:49:40, Table 2) are kept by the cloud provider in the project's job history, which only members of the project can read; we copied them into the data description committed in `7054746` (23:24:39 UTC on 9 October). Outsiders can therefore check the content of the registration, the hashes and the order of the commits, but the claim that the registration preceded the W1 extraction rests on the commit and job times, not on a third-party timestamp. A trusted timestamp, such as anchoring the commit hash on chain or in an independent registry before extraction, would close this gap.

D. THE MAIN ANALYSIS

After the registered results were known, the revised analysis was defined: amounts in US dollars on five tokens; scores from the twelve months before each freeze; graphs whose nodes are contracts other than smart accounts, with edges read from the whole chain; each allowance counted up to its owner's balance at the freeze, with balances reconstructed from `Transfer` logs; constants set on the contract graph; and the discovery-order test. This revised analysis is the main analysis of the paper, and we call it the contract-graph analysis; the discovery-order test is reported in the companion manuscript [15].

Each definition has a reason in what the scores are meant

TABLE 3. Commit Identifiers of the Registered Record (Times in UTC) and of the Main Analysis, and SHA-256 Hashes of the Registration Files and the Plan

Plan and configuration (rule A), 9 Oct 13:41:27; `43d8045` in the authors' repository
`d060f3a68c15ed22bc2a566cfb83f0a546ca85b3`
Registration of W1 (rule B), 9 Oct 16:49:32; `0d3a057` in the authors' repository
`1747e64884a2fd0adf1f76433e604474387d5ce2`
Registered results committed, 9 Oct 18:00:55; `8d8b568` in the authors' repository
`344017eba74cc80350071f6e6967db1e5dbf7667`
Registered benchmark and data description, 9 Oct 23:24:39; `f33a87a` in the authors' repository
`7054746eb0eb9cabbc61b12bbfbef8dff9e01071`
Main analysis (contract graph), defined after the registered results were known
Published results: summaries, digest and checksums
`6b389df1b5718a876e8fd3e07d975157d32a5773`
Code revision at which they were computed
`bb78da330d39a430c54922ba41c6da9e98e02e5d`
List of the per-contract files in the public Drive folder
`7aa792847f72f55a0741a66d8454e27364d84aa3`
SHA-256 of the registration document, CRLF line ends (recorded in the manifest)
`2126c8dc3fe3c0f9273aa4ecfc1e15bd`
`dc1f5a6f65e98548b9528f2b18ca2a6b`
SHA-256 of the registration document, LF line ends (as git stores it)
`34cd6dfeb56b8fb8a553cb24afbff06c`
`6a94e6f69a2eb7f7d642b0cc39e54b2c`
SHA-256 of the configuration, CRLF line ends (recorded in the manifest)
`d6a30adbcabcf511d6892332da73639e`
`fdbece25891b392e13e90faf09b7a833`
SHA-256 of the configuration, LF line ends (as git stores it)
`2ea235021241fa408f8776d55c2b594d`
`34ced0ca1dad44e16f22775c5db8297f`
SHA-256 of the plan in `d060f3a`, LF line ends
`7ec80d49c17331bd10e086fecd909a3f`
`36020dbb284f0dbfbb81a8549f84fd9f`

Identifiers are those of the public repository. For the registered record the table also names the commit of the authors' Git repository from which each was copied with its times and contents, as the extraction manifest, the registered digest and the repository's HISTORY.md record it. The commits of the main analysis carry no time here: none of them preceded the registered results. Each SHA-256 value is printed over two lines; join them without a space.

to rank. Raw base units cannot be summed over tokens, since a USDC has 10^6 base units and a wrapped ether 10^{18}, and with the plan's $b = 1$ on raw base units $V(z)$ is within 10^{-4} of one for 99.52% of the transfers and 99.78% of the latest allowances of the registered analysis; amounts are therefore valued in US dollars, on five tokens with reliable prices. One year per freeze scores both freezes on windows of equal length and keeps events older than a year out of every score. Nothing has to be deployed for a user account, and who controls one is not measured here; user accounts also hold most of the allowances toward a contract (Section V-F), so in a graph of all accounts a walk is largely a walk over them. With contracts as the only nodes every node costs a deployment, and every edge records a relation that one contract holds toward another. An allowance beyond the owner's balance cannot be spent, and about two thirds of the positive latest allowances between contracts are unlimited (Table 4), so each allowance counts up to the balance. The constants of the weights are quantiles of the events between contracts in the twelve months up to the W0 freeze, so that no later event enters them.

Rule A and the criteria F1 and F2 of Rule B are applied to the scores of the contract graph with their thresholds, δ and λ. The main analysis is a deviation from the plan and the registration, defined with the registered results in view and with the labels of both windows known. Every verdict on its scores is a re-analysis, and none is a registered result. Table 3 identifies its published results and the code revision that computed them.

E. WHAT WAS NOT PRE-SPECIFIED

Nothing in the main analysis was pre-specified: its definitions and every number computed from them came after the registered results. What it takes from the plan and the registration are the rules, the margin, the definitions of the scores as far as Section IV-D leaves them, the definitions of the labels and the bootstrap. In the registered analysis every contrast outside A1, A2, F1, F2, F6, S1 and S2 was computed after the labels of its window were known: the comparison of C-PR with the payment walk on new approval pairs in W0, the S-PR contrasts, every contrast against a raw count, and every comparison of C-PR with EndorseRank or AWP outside S1. The plan commit holds the plan, the configuration, the query that copies the logs and the BigQuery runner. The queries and scripts that build the cohort, the extraction around it, the pair tables and the W0 labels, the implementations of every score and of the bootstrap, and the drivers of the same-window and W0 evaluations entered the repository with the registration commit, after they had run on the observation window and after Rule A had been judged, so their text before that point is not on record; the unit tests of Section III entered with them. The W1 extraction and evaluation used them as committed. Every check in Appendix A was run after all of the above.

V. DATA AND MEASUREMENT

A. SOURCES AND EVENTS

All logs come from Arbitrum One (chain ID 42161), an optimistic rollup on Ethereum [1], [43], through the public BigQuery table

`bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`.

Scan A copied the logs of the observation window, 1 October 2023 to 30 June 2026, and scan B those of July to September 2026. Approval and transfer logs are selected by their ERC-20 event signatures and exactly three topics, which excludes ERC-721 tokens.

The main analysis reads from these copies every `Approval`, of any value, and every non-zero `Transfer` of five tokens from 1 April 2025 to 30 September 2026, wherever on the chain they occurred: 1,240,275,042 rows that name 25,537,280 distinct addresses other than the zero address. A zero-value approval is read because it deletes an allowance. The five tokens, WETH, USDC, USDT0, ARB

and WBTC, are those with the most rows in the twelve months before 30 June 2026 among the tokens priced by a Chainlink market-price feed of low or medium market risk [44]. Every amount is valued in US dollars at the daily close of the DefiLlama price series [45], a transfer at the price of its day and an allowance at the price of the freeze day. Each score uses the twelve months that end at its freeze: July 2025 through June 2026 for W1 and the same-window checks, April 2025 through March 2026 for W0. Of the rows, 603,475,784 have a contract at both ends; only these can build an edge.

The registered analysis scores every event of the observation window up to its freeze, in raw base units and on every ERC-20 token, and it reads only the events that name a contract of its cohort as owner, spender, sender or recipient: 389,866,882 approval and 1,538,404,067 transfer logs.

B. ACCOUNT TYPES

An address is a user account if it sent a transaction, if it holds no code, if its code is an EIP-7702 delegation designator [46], or if it is a smart account, by which we mean a Safe [47] or an ERC-4337 account [48]. All other addresses that hold code are contracts, and only they are nodes.

Of the 25,537,280 addresses, 17,477,779 sent a transaction and are user accounts by that fact; apart from an EIP-7702 delegation, an account with code cannot send one (EIP-3607). We read the code of 6,230,270 addresses with `eth_getCode` at block 513,216,506, the block at which the plan's cohort candidates had been typed: every address that sent no transaction and has some counterparty other than a transaction sender or the zero address, the members of the cohorts, and two audit samples of 1,000 addresses each. An address outside this set that sent no transaction has only user accounts and the zero address as counterparties and cannot end an edge between contracts. Of the addresses read, 2,909,509 hold no code and 1,165,025 an EIP-7702 delegation. A candidate contract is a Safe when `getThreshold()` answers a value from 1 to 1,000 and `getOwners()` a non-empty list (46,204 addresses). It is an ERC-4337 account when `entryPoint()` answers a canonical EntryPoint and its code, or its implementation's code for a proxy, carries a `validateUserOp` selector and no paymaster selector (87,157). Accounts that this test misses because a proxy hides the selector, or because the getter has another name, are typed from listed code families, 20 code hashes and 7 pairs of code hash and implementation whose accounts answer a canonical EntryPoint; they add 192,656 ERC-4337 accounts. The remaining 1,829,719 addresses are contracts, against 21,877,279 typed user accounts. No call was aborted at every attempt, and none of the 1,000 audited transaction senders held code. A kind of smart account that the rule does not recognize stays a contract (Section VIII).

C. BALANCES AND THE BALANCE CAP

The balance of an owner in a token at a freeze is the net flow of the token's `Transfer` logs to the owner up to the freeze, from the public logs before the observation window and the copied logs since. A value that is negative or below 10^{-6} token counts as zero. To check the reconstruction we sampled 1,500 pairs of a token and an owner, for each token 150 with a zero and 150 with a positive balance, and compared them with `balanceOf` at the last block of 30 June 2026: 1,493 agree (99.5%; the threshold set for the check was 98%).

An allowance counts up to this balance. Of the 339,282 positive latest allowances between contracts at the freeze of 30 June 2026, 223,892 (66.0%) are unlimited. The cap sets 222,559 (65.6%) to zero, since their owners hold nothing of the token, and leaves 116,723 positive, 114,454 of them at less than the approved amount (Table 4). At the freeze of 31 March 2026 it sets 93.6% of 308,922 to zero. Most allowances that one contract holds over another are thus backed by no balance at the freeze.

D. CONSTANTS

The constants are set on the contract graph from the twelve months up to the W0 freeze and rounded to two significant digits. The median dollar value of a transfer between contracts gives $m_T = 160$ dollars (164.05 before rounding), and the 99th percentile of the same values gives $U = 16{,}000$ dollars (16,412). The median positive capped allowance between contracts at the W0 freeze, unlimited allowances included, gives $m_A = 0.071$ dollars: half of the allowances that survive the cap are worth less than eight cents. EndorseRank's value weight therefore saturates early, at $V = 1.000$ for a capped allowance of one dollar.

E. GRAPHS

Table 4 gives the graph at both freezes. At the freeze of 30 June 2026 the authorization layer has 115,006 pairs from 9,459 owner contracts to 100,112 spender contracts, and the payment layer 2,580,709 pairs. The union graph joins the two layers, without the pairs of a contract with itself, and counts 2,575,632 pairs among 1,299,438 contracts. The authorization layer between contracts is thin, and at this freeze one owner contract holds 98,073 of its 115,006 pairs; at the freeze of 31 March 2026 the layer has 17,940 pairs, and the owner with the most holds 256. We did not identify that contract (Section VIII). The walks use the pairs of positive weight: for a few dust pairs the bounded weight V evaluates to zero in floating point, which leaves 114,997 authorization pairs for EndorseRank and 2,580,381 payment pairs for AWP at the freeze of 30 June 2026, while the amount-weighted layers of C-PR and S-PR keep those pairs (Table 1).

F. COHORTS, COVERAGE AND LABELS

The matched cohort holds every contract to which at least three distinct owners gave a non-zero approval of one of the five tokens within the score window. The rule counts every approving owner, whatever its type; the cohort member itself is a contract. Of the 6,483 addresses with code that meet the rule in the twelve months before 30 June 2026, 40 are smart accounts and 6,443 are contracts, the matched cohort;

TABLE 4. The Contract Graph of the Main Analysis at the Two Freezes

Freeze	31 March 2026	30 June 2026
Latest allowances between contracts (token, owner, spender)		
Positive latest allowances	308,922	339,282
unlimited	208,245	223,892
share	67.4%	66.0%
capped to zero (no balance held)	289,299	222,559
share	93.6%	65.6%
capped below the allowance	17,480	114,454
positive after the cap	19,623	116,723
Authorization layer (owner, spender)		
Pairs	17,940	115,006
owner contracts	10,153	9,459
spender contracts	2,050	100,112
pairs of the largest owner	256	98,073
Pairs without the cap (sensitivity)	275,917	302,559
Payment layer (sender, recipient)		
Pairs	2,269,073	2,580,709
sender contracts	873,388	1,007,310
recipient contracts	626,356	801,813
Union graph (either layer, two different contracts)		
Pairs	2,269,421	2,575,632
Contracts	1,111,554	1,299,438

Every end of every pair is a contract; no user account is a node. An allowance is the latest `Approval` of a (token, owner, spender) triple in the score window, kept when positive and counted up to the owner's balance of the token at the freeze; a pair sums two contracts over the five tokens. Constants set on this graph: $m_T = 160$, $m_A = 0.071$ and $U = 16{,}000$ dollars (Section V-D).

TABLE 5. The Cohorts in the Contract Graph, and What Their Labels Count (Main Analysis)

(a) Cohort	Contracts	In the union graph		In the authorization layer	In no pair
W0 spenders	7,864	5,700	72.5%	24.2%	27.5%
W1 spenders	7,915	5,759	72.8%	25.0%	27.2%
Matched cohort	6,443	4,270	66.3%	19.4%	33.7%

(b) Labels and standing allowances	W0 spenders	W1 spenders
New approval pairs in the label window	748,258	702,275
from user accounts	90.4%	87.9%
spenders that gained one	1,725	1,491
New transfer senders in the label window	651,474	618,461
from user accounts	86.3%	82.3%
spenders that gained one	1,416	1,369
Positive allowance pairs at the freeze	4,116,471	3,809,524
owner is a user account	3,926,375	3,611,886
owner is a contract	190,096	197,638

(a) W0 spenders at the freeze of 31 March 2026; W1 spenders and the matched cohort at that of 30 June 2026. A contract in no pair between two different contracts lies outside the union graph and scores zero under every score and count. (b) The labels count every account, user accounts included, which no score reads. The lower block counts, without the balance cap, the pairs of an owner and a spender of the cohort with a positive latest allowance at the freeze, the pairs against which a new approval pair is new.

at the W0 freeze 6,292 of 6,343 are contracts. The plan's cohort counts the approvals of every token in the observation window: of the 29,022 spender addresses that met the rule, 27,844 held contract code at the same block, while 1,121 held none and 57 held an EIP-7702 delegation designator.

The spender cohort of a window holds every contract with a positive latest allowance at the freeze in the approvals of the score window that name an address meeting the cohort rule as owner or spender; it includes contracts outside the matched cohort that such an address approved. Of the 7,962 spenders at the W0 freeze, 7,864 are contracts (97 are user accounts and one is a system address), and of the 7,995 at the W1 freeze, 7,915 (79 and one). Their first label counts new approval pairs: owners that gave the spender a positive approval during the label window although the pair held no positive allowance at the freeze. Their second label counts new transfer senders: addresses other than the spender that sent it a transfer during the label window but had not done so between the start of the score window and the freeze. The labels are counted on the same events as the spender list, without the balance cap, and they count every account: 90.4% of the 748,258 new approval pairs of W0 and 87.9% of the 702,275 of W1 come from user accounts, as do 86.3% and 82.3% of the new senders (Table 5). The same holds for the allowances already in place: of the 3,809,524 pairs of an owner and a W1 spender with a positive allowance at the freeze, 3,611,886 have a user account as owner.

A cohort contract that ends no pair between two different contracts has no place in the union graph, and every score and count gives it zero. That is the case for 27.5% of the W0 spenders, 27.2% of the W1 spenders and 33.7% of the matched cohort, and only 24.2%, 25.0% and 19.4% of them are ends of an authorization pair (Table 5). The graph between contracts thus covers two thirds to three quarters of the contracts it is asked to rank, and its authorization layer a fifth to a quarter.

G. STATISTICS

Every coefficient is the tie-corrected Kendall τ_b [14], computed with SciPy, with a 95% percentile interval from a paired bootstrap over the contracts of the cohort [49], [50]. Resample indices are drawn with NumPy's default generator from seed 42, one index matrix per cohort, and the same matrix serves every score on a label, so each difference between two scores is a paired difference. Every result uses 400 resamples. The supplementary script recomputes the contrasts of the rules, except those of S2, from the published per-contract score files and reproduces the published values exactly (Appendix A).

VI. RESULTS

Each subsection reports the main analysis first, a re-analysis under the rules, and then the registered analysis, the test that the rules were fixed for; Tables 6 and 7 and Figs. 2 and 3 give both. Two facts frame every out-of-window number of the main analysis (Section V-F). Roughly 27% of each spender cohort has no edge in the union graph, so every score and count gives it zero, and a contrast over a whole cohort also reflects how two scores order the contracts with an edge against those without one. And the labels count mostly user accounts, which no score reads. The coefficients are accordingly small.

A. SAME WINDOW: RULE A'S SECONDARY PART FAILS

On the matched cohort, C-PR at $\lambda = 0.5$ lies between its two layers on every family. In the main analysis it reaches 0.738 $[0.729, 0.747]$ on the transfer family, below the payment walk's interval $[0.799, 0.809]$; 0.433 $[0.419, 0.445]$ on the authorization family, far below the authorization walk's interval $[0.827, 0.850]$; and 0.574 $[0.564, 0.585]$ on the Sybil-stability family, above the authorization walk (0.222 $[0.207, 0.238]$) but below the payment walk (0.637 $[0.628, 0.646]$). The registered scores give the same pattern: 0.296 $[0.289, 0.303]$ against $[0.522, 0.534]$, 0.418 $[0.412, 0.423]$ against $[0.506, 0.517]$, and 0.129 $[0.124, 0.134]$ between 0.020 $[0.015, 0.026]$ and 0.311 $[0.304, 0.317]$. All three parts of the secondary criterion therefore fail in both analyses. The paired contrasts agree: C-PR minus the payment walk on the transfer family is -0.066 $[-0.074, -0.058]$ in the main and -0.232 $[-0.239, -0.224]$ in the registered analysis, and C-PR minus the authorization walk on the authorization family -0.406 $[-0.418, -0.393]$ and -0.094 $[-0.099, -0.088]$. In the main analysis C-PR stays close to its payment layer, because few contracts have an authorization edge: 5,193 of the 6,443 cohort contracts are outside the authorization graph and score zero under the authorization walk. These checks come from the events that build the graphs, so they show which construct a score tracks and are not external evidence [41].

B. W0: RULE A'S PRIMARY PART FAILS

Table 6 gives every score against the later labels, Table 7 the contrasts, and Fig. 2 the contrasts that the rules fixed. In the main analysis both parts fail. A1 fails: on new approval pairs C-PR lies below the authorization walk (-0.025 $[-0.045, -0.004]$; 0.142 against 0.167), so it is worse, not "no worse". A2 fails: on new transfer senders C-PR lies below the payment walk (-0.014 $[-0.019, -0.008]$; 0.254 against 0.268). With its primary part failed, Rule A answers in the negative whether C-PR beats a walk over either layer alone. The registered analysis, the test that Rule A was written for, fails the primary part by A2 alone: A1 holds there ($+0.008$ $[0.003, 0.013]$) and A2 fails (-0.030 $[-0.034, -0.026]$). The upper end of the A1 interval in the main analysis lies close to zero, but its sign does not depend on the bootstrap seed (Appendix A).

A comparison that Rule A did not contain pointed the other way. In the registered analysis, the only one that existed when Rule B was written, C-PR ranked new approval pairs much better than the payment walk alone (0.236 against 0.142; $+0.094$ $[0.085, 0.103]$). Because this contrast was examined after the labels were known, we treated it as a hypothesis and wrote Rule B to test it on a window that this study had not yet extracted. The scores of the contract graph give $+0.071$ $[0.059, 0.084]$ for the same contrast.

C. W1: RULE B IS NOT MET

In the main analysis F1 holds: C-PR ranks the spenders that later gain approval pairs at $\tau = 0.104$ $[0.083, 0.126]$ against 0.030 $[0.008, 0.050]$ for the payment walk, a difference of $+0.074$ $[0.060, 0.087]$. F2 fails: on new transfer senders C-PR reaches 0.236 $[0.217, 0.255]$ against 0.253 $[0.232, 0.272]$, a difference of -0.016 $[-0.022, -0.012]$ whose whole interval lies beyond the margin of -0.01, so C-PR ranks later payers worse than the payment walk by more than the registration allowed. Rule B is not met. We do not claim that C-PR improves on the payment walk.

The registered analysis, the test that the registration fixed, gives the same readings of the two criteria (Table 7). F1 holds ($+0.101$ $[0.092, 0.111]$; 0.237 against 0.136) and F2 fails with its whole interval beyond the margin (-0.025 $[-0.029, -0.021]$; 0.235 against 0.260). Across 50 further bootstrap seeds no verdict of F1 or F2 changes in either analysis, and the F2 interval stays wholly beyond the margin in every seed (Appendix A).

D. RESULTS REPORTED BESIDE THE DECISION

These verdicts are nominal: each uses an unadjusted 95% interval, and none was conditioned on another. F6, Rule A's primary part rerun on W1, fails in both analyses. In the main analysis both of its parts fail, as in W0: C-PR lies below the authorization walk on new approval pairs (-0.043 $[-0.061, -0.024]$) and below the payment walk on new senders. In the registered analysis C-PR is no worse than the authorization walk on new approval pairs ($+0.010$ $[0.005, 0.014]$) but lies below the payment walk on new senders.

Neither sensitivity analysis changes a reading in either analysis. With AWP as the comparator (S1), F1 holds ($+0.040$ $[0.026, 0.055]$ in the main analysis; $+0.060$ $[0.050, 0.070]$ in the registered one) and F2 fails with its whole interval far beyond the margin: AWP ranks new senders at 0.314 $[0.295, 0.332]$ and 0.322 $[0.316, 0.329]$, and C-PR's deficit is -0.077 $[-0.086, -0.069]$ and -0.087 $[-0.093, -0.082]$. Against the published payment-based method, then, C-PR trades a measurable part of its ranking of later inflow for its gain on later approvals. With the contracts that a graph does not contain kept as isolated nodes (S2), no verdict changes: in the main analysis F1 is $+0.073$ $[0.059, 0.086]$ and F2 -0.016 $[-0.022, -0.011]$. F6a moves from -0.043 to -0.062 $[-0.081, -0.041]$, because the isolated-node rule lifts the spenders outside the authorization graph from zero to the score of a node without in-edges, which raises the authorization walk's coefficient on new approval pairs from 0.147 to 0.173.

E. INSIDE THE UNION GRAPH, AND WITHOUT THE BALANCE CAP

Table 8 repeats the contrasts of the rules in two settings that decide no rule. Among the spenders inside the union graph, 5,700 in W0 and 5,759 in W1, the contrasts with the payment walk keep their sign and grow: F1 is $+0.118$ $[0.101, 0.138]$, F2 -0.020 $[-0.027, -0.011]$ and A2 -0.016 $[-0.025, -0.007]$. The contrasts with the authorization walk do not exclude zero there: A1 is $+0.020$ $[-0.002, 0.043]$ and

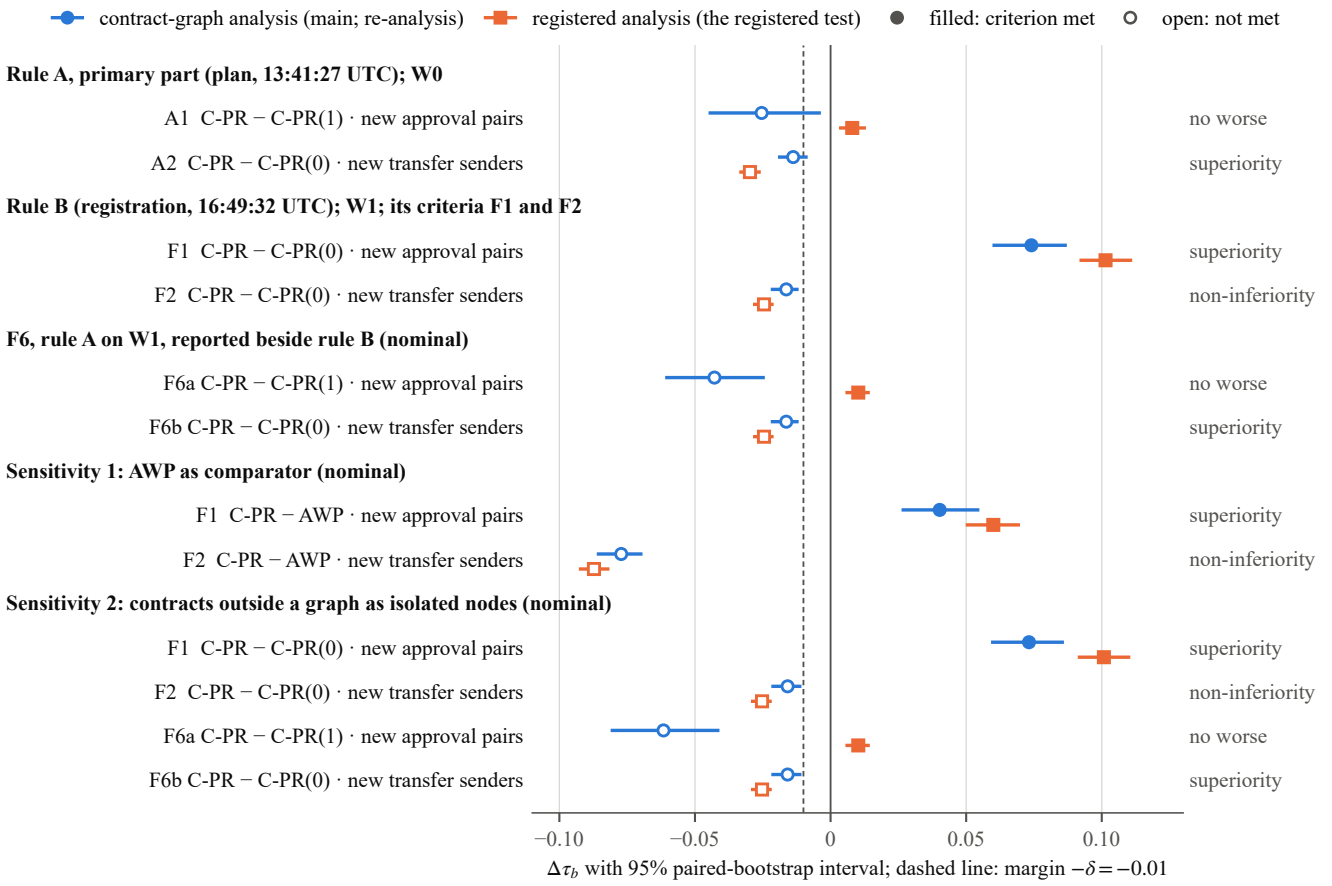


FIGURE 2. **Contrasts fixed by Rule A, the two applied criteria of Rule B (F1, F2), F6 and the two registered sensitivity analyses, with 95% paired-bootstrap intervals (400 resamples, seed 42). Circles: contract-graph analysis, the main analysis, defined after the registered results were known and a re-analysis under the same rules; squares: registered analysis, the test that the rules were fixed for. Filled markers: criterion met; open markers: not met. The dashed line is the margin $-\delta = -0.01$. Only A1, A2, F1 and F2 bear on a rule; the other verdicts are nominal. No value of the main analysis is a registered result. Values in Table 7.**

F6a $+0.007$ $[-0.017, 0.029]$, and across 50 further seeds the A1 interval contains zero in 49 (Appendix A). On the whole cohorts C-PR ranks every spender that the payment layer reaches above the spenders without any edge between contracts, whereas the authorization walk leaves both groups tied at zero. That ordering does not help on this label: the 2,421 W0 spenders without a payment edge gained a new approval pair as often as the others (23.4% against 21.3%), although they gained a new sender far less often (7.9% against 22.5%). The failure of A1 on the whole cohort is thus consistent with the place that C-PR gives to the spenders outside the union graph, and it does not show a worse order among the spenders inside it.

Without the balance cap, every positive latest allowance between contracts counts at its own value, and the authorization layer has 275,917 pairs at the W0 freeze where the cap leaves 17,940. The authorization walk alone then ranks new approval pairs much lower (0.086 against 0.167 in W0), and C-PR lies above it ($+0.064$ $[0.051, 0.076]$ in W0; $+0.041$ $[0.031, 0.053]$ in W1), so that A1 and F6a would hold. A2 and F2 fail as before (-0.019 $[-0.027, -0.012]$; -0.018 $[-0.028, -0.011]$), and F1 holds ($+0.088$ $[0.074, 0.103]$). The balance cap thus makes the authorization walk a stronger comparator. It does not change that C-PR falls below the payment walk on later payers.

F. THE WEIGHT SWEEP

Fig. 3 plots each label against λ. On new approval pairs the first step is the large one: switching the authorization layer on lifts C-PR from 0.030 to 0.103 in the W1 of the main analysis and from 0.136 to 0.235 in the registered W1, and the interior values differ by less than 0.003. On new transfer senders the interior curve falls slowly as λ rises (0.239, 0.236, 0.234 in the main W1; 0.240, 0.235, 0.228 in the registered one), and every interior value lies below the payment walk. The two analyses differ at the upper end. In the main analysis the authorization walk ranks new approval pairs above every interior value (0.147 against 0.104 in W1), so the interior of the family lies below both end points, each on its own label; in the registered analysis the interior lay above the authorization walk (0.237 against 0.227). In both, the authorization walk falls far on new senders (0.064 and 0.133), because at $\lambda = 1$ the payment rows

TABLE 6. Kendall τ_b Between Scores at the Freeze and Later Labels, With 95% Paired-Bootstrap Intervals

Contract-graph analysis (the main analysis; a re-analysis defined after the registered results were known)				
	W0 spenders ($n = 7{,}864$)		W1 spenders ($n = 7{,}915$)	
Score	New approval pairs	New transfer senders	New approval pairs	New transfer senders
In-approve degree at the freeze	0.204 [0.181, 0.228]	0.091 [0.066, 0.113]	0.181 [0.158, 0.208]	0.072 [0.046, 0.096]
In-approve value at the freeze	0.189 [0.166, 0.213]	0.081 [0.057, 0.103]	0.163 [0.141, 0.188]	0.062 [0.037, 0.084]
Transfer in-degree at the freeze	0.070 [0.050, 0.091]	0.318 [0.298, 0.336]	0.031 [0.008, 0.052]	0.302 [0.278, 0.323]
Union in-degree at the freeze	0.126 [0.108, 0.146]	0.306 [0.287, 0.324]	0.093 [0.071, 0.114]	0.292 [0.270, 0.312]
EndorseRank	0.168 [0.145, 0.193]	0.084 [0.062, 0.105]	0.147 [0.125, 0.170]	0.064 [0.041, 0.085]
EndorseRank with activity restarts	0.163 [0.142, 0.188]	0.089 [0.069, 0.111]	0.148 [0.126, 0.171]	0.072 [0.049, 0.093]
AWP	0.098 [0.079, 0.119]	0.330 [0.313, 0.347]	0.064 [0.042, 0.084]	0.314 [0.295, 0.332]
C-PR ($\lambda = 1$), authorization walk	0.167 [0.145, 0.192]	0.084 [0.062, 0.105]	0.147 [0.124, 0.170]	0.064 [0.041, 0.084]
C-PR ($\lambda = 0.75$)	0.142 [0.124, 0.162]	0.251 [0.232, 0.267]	0.104 [0.082, 0.126]	0.234 [0.215, 0.252]
C-PR ($\lambda = 0.5$), primary	0.142 [0.124, 0.162]	0.254 [0.235, 0.270]	0.104 [0.083, 0.126]	0.236 [0.217, 0.255]
C-PR ($\lambda = 0.25$)	0.141 [0.123, 0.161]	0.256 [0.238, 0.273]	0.103 [0.081, 0.124]	0.239 [0.219, 0.258]
C-PR ($\lambda = 0$), payment walk	0.070 [0.052, 0.091]	0.268 [0.250, 0.285]	0.030 [0.008, 0.050]	0.253 [0.232, 0.272]
S-PR	0.091 [0.074, 0.113]	0.264 [0.246, 0.283]	0.055 [0.034, 0.073]	0.242 [0.222, 0.261]
Registered analysis (the test that the rules were fixed for)				
	W0 spenders ($n = 33{,}101$)		W1 spenders ($n = 34{,}587$)	
Score	New approval pairs	New transfer senders	New approval pairs	New transfer senders
In-approve degree at the freeze	0.252 [0.244, 0.260]	0.156 [0.148, 0.166]	0.262 [0.255, 0.269]	0.179 [0.171, 0.186]
Transfer in-degree at the freeze	0.148 [0.138, 0.158]	0.302 [0.294, 0.310]	0.145 [0.135, 0.155]	0.303 [0.296, 0.309]
EndorseRank	0.241 [0.234, 0.248]	0.226 [0.219, 0.234]	0.233 [0.226, 0.240]	0.210 [0.203, 0.217]
EndorseRank with activity restarts	0.295 [0.288, 0.302]	0.272 [0.265, 0.280]	0.280 [0.272, 0.287]	0.245 [0.237, 0.252]
AWP	0.196 [0.187, 0.206]	0.340 [0.333, 0.348]	0.177 [0.166, 0.187]	0.322 [0.316, 0.329]
C-PR ($\lambda = 1$), authorization walk	0.228 [0.219, 0.236]	0.122 [0.113, 0.133]	0.227 [0.219, 0.234]	0.133 [0.125, 0.142]
C-PR ($\lambda = 0.75$)	0.236 [0.229, 0.243]	0.217 [0.209, 0.226]	0.237 [0.230, 0.244]	0.228 [0.220, 0.236]
C-PR ($\lambda = 0.5$), primary	0.236 [0.229, 0.243]	0.224 [0.216, 0.234]	0.237 [0.230, 0.244]	0.235 [0.227, 0.243]
C-PR ($\lambda = 0.25$)	0.234 [0.227, 0.241]	0.230 [0.222, 0.240]	0.235 [0.228, 0.242]	0.240 [0.232, 0.247]
C-PR ($\lambda = 0$), payment walk	0.142 [0.133, 0.151]	0.254 [0.246, 0.262]	0.136 [0.125, 0.145]	0.260 [0.252, 0.266]
S-PR	0.177 [0.167, 0.186]	0.245 [0.236, 0.253]	0.169 [0.159, 0.178]	0.251 [0.244, 0.258]

W0: scores frozen on 31 March 2026, labels counted from April to June 2026. W1: scores frozen on 30 June 2026, labels counted from July to September 2026. In the main analysis every score is computed from the twelve months before its freeze on the graph of contracts (amounts in US dollars on five tokens, allowances counted up to the owner's balance), and the counts are taken on that graph: owner contracts with a capped allowance to the contract (in-approve degree), the summed capped value of those allowances (in-approve value), other contracts that sent it a transfer (transfer in-degree), and other contracts with an edge to it in either layer (union in-degree). 5,700 of the W0 spenders (72.5%) and 5,759 of the W1 spenders (72.8%) lie in the union graph; the others score zero in every row of the upper block. 1,725 W0 spenders gained a new approval pair and 1,416 a new transfer sender, and 1,491 and 1,369 W1 spenders did; in the registered analysis 3,403, 3,042, 2,850 and 2,812. In the registered analysis every score is computed in raw base units from every extracted event of every ERC-20 token since 1 October 2023, and the two counts are taken over all accounts. No value of the upper block is a registered result. The count, EndorseRank and AWP rows are reference scores that the companion study [15] also reports. Intervals: 400 paired resamples over the contracts of the cohort, seed 42.

of contracts without authorization out-edges are no part of the walk. The F1 gain over the payment walk is thus a property of switching the authorization layer on, not of the value $\lambda = 0.5$, and no interior value recovers the payment walk's ranking of later payers. Appendix A gives the paired steps from $\lambda = 0$ to 0.25, from 0.25 to 0.75 and from 0.75 to 1, and the rank agreement between these values.

G. SEEDED RESTARTS

S-PR stays close to the payment walk on new senders and far below the authorization walk on new approval pairs. In W0, S-PR minus the authorization walk on new approval pairs is -0.077 $[-0.098, -0.055]$ in the main and -0.051 $[-0.060, -0.043]$ in the registered analysis, and S-PR minus the payment walk on new senders is -0.003 $[-0.013, 0.008]$ and -0.009 $[-0.012, -0.006]$. Its restarts lift its coefficient on new approval pairs only part of the way from the payment walk toward the authorization walk (0.091 against 0.070 and 0.167 in the main W0; 0.177 against 0.142 and 0.228 in the registered one; no paired contrast with the payment walk was computed). With $d = 0.85$ only 15% of the mass, plus the dangling mass, passes through the restart vector. In contrasts computed after the labels were known, C-PR ranks new approval pairs above S-PR in both windows of the main analysis ($+0.051$ $[0.038, 0.064]$ and $+0.049$ $[0.036, 0.063]$); on new senders it lies slightly below S-PR in W0 (-0.011 $[-0.021, -0.0004]$) and level with it in W1 (-0.006 $[-0.015, 0.002]$). Authorization edges thus help the walk on later approvals more when they carry it, fully in the authorization walk and mixed in C-PR, than when they only decide where it restarts.

H. WALKS AGAINST COUNTS

On new approval pairs no walk ranked the spenders as well as the in-approve degree, in either analysis. In the main analysis EndorseRank falls short of it by -0.036 $[-0.046, -0.027]$ in W0 and -0.034 $[-0.043, -0.025]$ in W1, EndorseRank with activity restarts by -0.041 $[-0.053, -0.029]$ and -0.033 $[-0.044, -0.022]$, the authorization walk by -0.037 $[-0.046, -0.028]$ in W0, and C-PR by -0.062

TABLE 7. Paired Contrasts in Both Analyses: Rule A, the Two Applied Criteria of Rule B, the Reports and Sensitivity Analyses Registered With Them, and Contrasts Computed After the Labels Were Known

		Contract-graph analysis (re-analysis)		Registered analysis (the test)	
Contrast (a minus b) and label	Test	$\Delta\tau$ [95% CI]	Verdict	$\Delta\tau$ [95% CI]	Verdict
Rule A, fixed in the plan (d060f3a) before any log of this study was extracted; W0 spenders					
A1: C-PR − C-PR ($\lambda = 1$), new approval pairs	no worse	−0.025 [−0.045, −0.004]	fails	+0.008 [0.003, 0.013]	holds
A2: C-PR − C-PR ($\lambda = 0$), new transfer senders	superiority	−0.014 [−0.019, −0.008]	fails	−0.030 [−0.034, −0.026]	fails
Rule A, primary part (A1 and A2)			**not met**		**not met**
Rule A, secondary part (same window)			**not met**		**not met**
Rule B, registered (1747e64) before this study extracted the W1 logs: two of its registered criteria, F1 and F2; margin $\delta = 0.01$; W1 spenders					
F1: C-PR − C-PR ($\lambda = 0$), new approval pairs	superiority	+0.074 [0.060, 0.087]	holds	+0.101 [0.092, 0.111]	holds
F2: C-PR − C-PR ($\lambda = 0$), new transfer senders	non-inferiority	−0.016 [−0.022, −0.012]	fails	−0.025 [−0.029, −0.021]	fails
Rule B, by F1 and F2			**not met**		**not met**
F6, the primary part of rule A on W1, registered to be reported beside the decision (nominal)					
F6a: C-PR − C-PR ($\lambda = 1$), new approval pairs	no worse	−0.043 [−0.061, −0.024]	fails	+0.010 [0.005, 0.014]	holds
F6b: C-PR − C-PR ($\lambda = 0$), new transfer senders	superiority	−0.016 [−0.022, −0.012]	fails	−0.025 [−0.029, −0.021]	fails
Registered sensitivity analysis S1: AWP as the comparator (nominal)					
F1: C-PR − AWP, new approval pairs	superiority	+0.040 [0.026, 0.055]	holds	+0.060 [0.050, 0.070]	holds
F2: C-PR − AWP, new transfer senders	non-inferiority	−0.077 [−0.086, −0.069]	fails	−0.087 [−0.093, −0.082]	fails
S1, by F1 and F2			**not met**		**not met**
Registered sensitivity analysis S2: contracts outside a graph kept as isolated nodes (nominal)					
F1: C-PR − C-PR ($\lambda = 0$), new approval pairs	superiority	+0.073 [0.059, 0.086]	holds	+0.101 [0.091, 0.111]	holds
F2: C-PR − C-PR ($\lambda = 0$), new transfer senders	non-inferiority	−0.016 [−0.022, −0.011]	fails	−0.025 [−0.029, −0.022]	fails
F6a: C-PR − C-PR ($\lambda = 1$), new approval pairs	no worse	−0.062 [−0.081, −0.041]	fails	+0.010 [0.005, 0.014]	holds
F6b: C-PR − C-PR ($\lambda = 0$), new transfer senders	superiority	−0.016 [−0.022, −0.011]	fails	−0.025 [−0.029, −0.022]	fails
S2, by F1 and F2			**not met**		**not met**
Computed after the labels of the window were known; no rule					
W0: C-PR − C-PR ($\lambda = 0$), new approval pairs	—	+0.071 [0.059, 0.084]		+0.094 [0.085, 0.103]	
W0: S-PR − C-PR ($\lambda = 1$), new approval pairs	—	−0.077 [−0.098, −0.055]		−0.051 [−0.060, −0.043]	
W0: S-PR − C-PR ($\lambda = 0$), new transfer senders	—	−0.003 [−0.013, 0.008]		−0.009 [−0.012, −0.006]	
W0: C-PR − AWP, new transfer senders	—	−0.077 [−0.087, −0.068]		−0.116 [−0.122, −0.109]	
W0: C-PR ($\lambda = 1$) − in-approve degree, new approval pairs	—	−0.037 [−0.046, −0.028]		−0.025 [−0.030, −0.019]	
W0: EndorseRank − in-approve degree, new approval pairs	—	−0.036 [−0.046, −0.027]		−0.011 [−0.016, −0.006]	
W0: AWP − transfer in-degree, new transfer senders	—	+0.012 [0.003, 0.023]		+0.038 [0.035, 0.042]	
W1: EndorseRank − in-approve degree, new approval pairs	—	−0.034 [−0.043, −0.025]		−0.029 [−0.032, −0.025]	
W1: AWP − transfer in-degree, new transfer senders	—	+0.011 [0.001, 0.022]		+0.020 [0.016, 0.023]	
W1: C-PR − EndorseRank, new approval pairs	—	−0.043 [−0.062, −0.025]		+0.004 [−0.001, 0.008]	

$\Delta\tau = \tau_a - \tau_b$ on the same resamples. C-PR without a λ is the primary setting $\lambda = 0.5$. Superiority holds when the interval lies above zero, non-inferiority when its lower end lies above $-\delta = -0.01$, and "no worse" when the interval contains zero or lies above it. This paper applies two of the registered criteria of rule B, F1 and F2; under the registration the rule holds only if each of its criteria holds, so a failure of either gives "not met". The left column, the main analysis, applies the rules and thresholds of the plan and the registration to scores defined after the registered results were known (graphs of contracts, amounts in US dollars on five tokens, scores from the twelve months before each freeze, allowances counted up to the owner's balance), so none of its verdicts is a registered result; the right column is the test that the plan and the registration fixed. Nominal verdicts use unadjusted 95% intervals. The count, EndorseRank and AWP rows are reference contrasts that the companion study [15] also reports. Intervals: 400 paired resamples, seed 42.

$[-0.082, -0.037]$ and -0.077 $[-0.097, -0.057]$ (Table 9). In the registered analysis EndorseRank falls short by -0.011 $[-0.016, -0.006]$ and -0.029 $[-0.032, -0.025]$, and the authorization walk by -0.025 $[-0.030, -0.019]$ in W0.

On new transfer senders the result has two sides. AWP ranks them slightly above the transfer in-degree on the whole cohorts ($+0.012$ $[0.003, 0.023]$ in W0 and $+0.011$ $[0.001, 0.022]$ in W1; $+0.038$ $[0.035, 0.042]$ and $+0.020$ $[0.016, 0.023]$ in the registered analysis). Among the 4,569 W0 spenders of the main analysis that another contract had paid before the freeze, the spenders AWP can order, it ranks them below the in-degree (-0.035 $[-0.049, -0.022]$); that subset was chosen after the results were known. C-PR ranks new senders below the transfer in-degree in both windows (-0.064 $[-0.075, -0.054]$ and -0.066 $[-0.075, -0.057]$).

The union in-degree counts the edges that C-PR walks. C-PR lies slightly above it on new approval pairs ($+0.016$ $[0.007, 0.024]$ in W0; $+0.011$ $[0.003, 0.019]$ in W1) and below it on new senders (-0.053 $[-0.061, -0.043]$; -0.056 $[-0.065, -0.047]$). Inside the union graph the directions are the same with larger gaps, and without the balance cap the lead on approval pairs is not resolved (Table 8). On each label, then, a count of one edge type ranks the spenders better than the coupled walk, and the walk improves on the count of both edge types only on later approvals and by less than 0.02.

The companion study reports the reference comparisons in full and adds a test of a different design, in which a crawl of the contract graph chooses the next contract by a score computed on the contracts read so far [15]. Its verdict on C-PR has the same shape: "Ordered by C-PR on the contracts read so far, a crawl from a random contract of the giant strongly connected component reached the contracts of the universe

TABLE 8. The Contrasts of the Rules and of C-PR With the Union In-Degree in Three Settings of the Main Analysis: $'\tau$ [95% CI]

Contrast (a minus b) and label	Whole spender cohort	Spenders inside the union graph	Whole cohort, no balance cap
W0, A1: C-PR − C-PR ($\lambda = 1$), new approval pairs	−0.025 [−0.045, −0.004]	+0.020 [−0.002, 0.043]	+0.064 [0.051, 0.076]
τ_a, τ_b	0.142, 0.167	0.221, 0.201	0.150, 0.086
W0, A2: C-PR − C-PR ($\lambda = 0$), new transfer senders	−0.014 [−0.019, −0.008]	−0.016 [−0.025, −0.007]	−0.019 [−0.027, −0.012]
W0: C-PR − union in-degree, new approval pairs	+0.016 [0.007, 0.024]	+0.035 [0.022, 0.049]	+0.004 [−0.004, 0.012]
W0: C-PR − union in-degree, new transfer senders	−0.053 [−0.061, −0.043]	−0.089 [−0.103, −0.075]	−0.037 [−0.045, −0.029]
W1, F1: C-PR − C-PR ($\lambda = 0$), new approval pairs	+0.074 [0.060, 0.087]	+0.118 [0.101, 0.138]	+0.088 [0.074, 0.103]
W1, F2: C-PR − C-PR ($\lambda = 0$), new transfer senders	−0.016 [−0.022, −0.012]	−0.020 [−0.027, −0.011]	−0.018 [−0.028, −0.011]
W1, F6a: C-PR − C-PR ($\lambda = 1$), new approval pairs	−0.043 [−0.061, −0.024]	+0.007 [−0.017, 0.029]	+0.041 [0.031, 0.053]
τ_a, τ_b	0.104, 0.147	0.204, 0.196	0.118, 0.077
W1: C-PR − union in-degree, new approval pairs	+0.011 [0.003, 0.019]	+0.027 [0.014, 0.041]	+0.0001 [−0.007, 0.008]
W1: C-PR − union in-degree, new transfer senders	−0.056 [−0.065, −0.047]	−0.093 [−0.107, −0.077]	−0.040 [−0.047, −0.031]

Left: the main analysis, as in Tables 7 and 9 (W0 n = 7,864, W1 n = 7,915). Middle: the same scores on the spenders that end a pair between two different contracts (5,700 in W0 and 5,759 in W1), with a bootstrap of that subset; a spender outside the union graph scores zero under every score and count, so on the whole cohort a contrast also reflects how the two scores order the spenders with a pair against those without one. Right: scores and the union in-degree recomputed with every positive latest allowance between contracts counted at its own value, which gives 275,917 authorization pairs at the W0 freeze where the cap leaves 17,940, and 302,559 where it leaves 115,006 at the W1 freeze. Under the two contrasts with the authorization walk, whose reading differs between the settings, the small figures give the two coefficients. The middle and right columns decide no rule. Intervals: 400 paired resamples, seed 42.

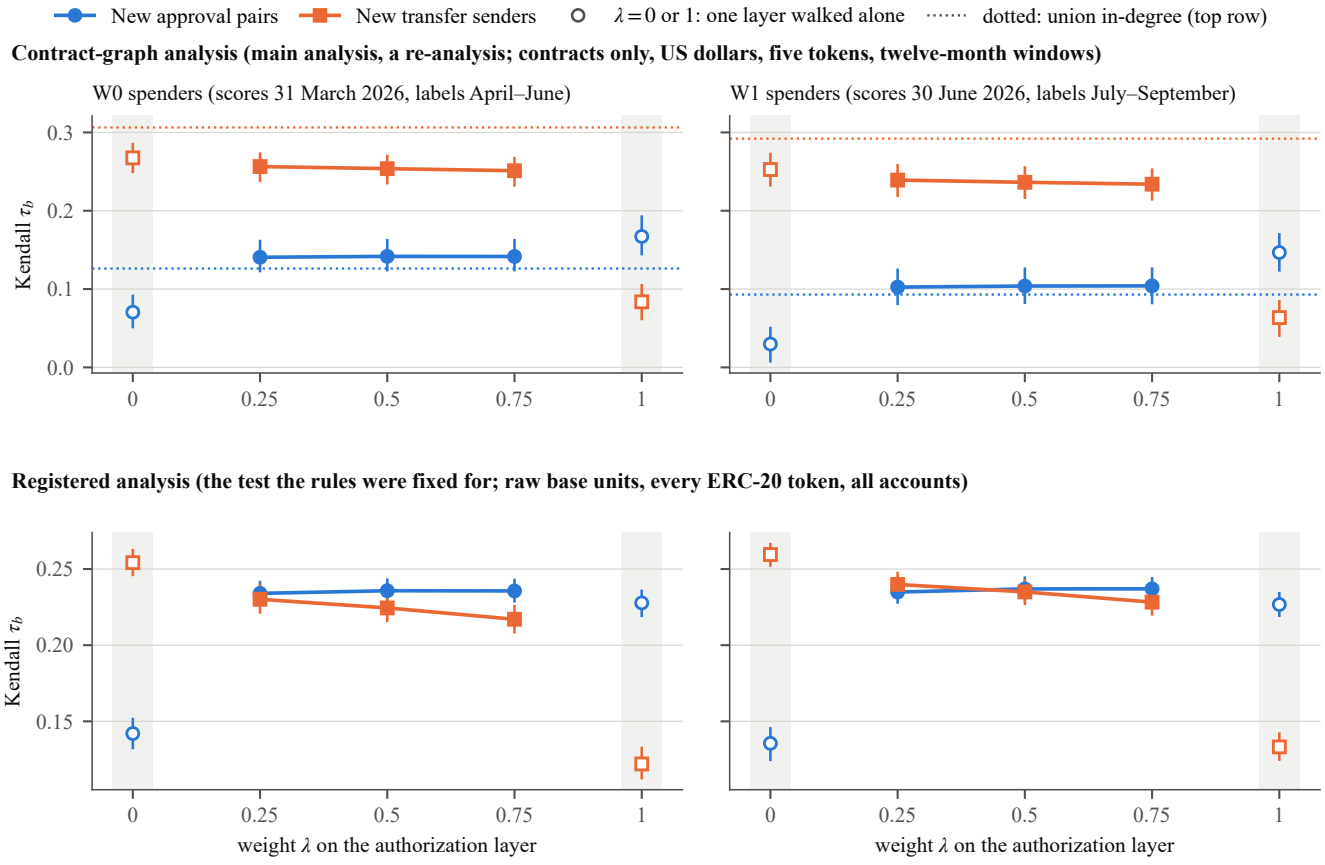


FIGURE 3. Kendall τ_b of C-PR with each later label against the authorization weight λ, with 95% paired-bootstrap intervals. Filled markers joined by lines: the grid $\lambda \in \{0.25, 0.5, 0.75\}$ fixed in the plan; open markers in the shaded bands: $\lambda = 0$ and $\lambda = 1$, the payment and authorization walks, separate operators that the line does not join. Top row: contract-graph analysis (the main analysis, a re-analysis; no value is a registered result), with the union in-degree, the count of the edges C-PR walks, as dotted lines; bottom row: registered analysis (the test that the rules were fixed for). Each row shares its vertical axis. Values in Table 6.

with the most user-account inflow and those with the most user-authorised value sooner than a breadth-first crawl, but not sooner than a crawl ordered by the number of contract backlinks. A crawl ordered by the number of contract backlinks reached them sooner." Proposition 2 accounts for the authorization side: for most spenders the walk is a reweighted owner count, and on this arrival label the reweighting did not help. The count results agree with preferential attachment

[26], [27] and with the strength of degree baselines in link prediction [28], [29].

I. AGAINST THE REFERENCE WALKS, AFTER THE LABELS

Rules A and B compare C-PR with its own layers, because only that contrast changes the coupling alone. Table 9 sets C-PR beside every other score of the main analysis; apart from the marked cells none of these contrasts belongs to a rule, and all were computed after the labels were known. Against EndorseRank, C-PR lies below on new approval pairs (-0.026 $[-0.046, -0.004]$ in W0; -0.043 $[-0.062, -0.025]$ in W1) and far above on new senders ($+0.170$ $[0.148, 0.191]$; $+0.172$ $[0.152, 0.193]$); in the registered analysis the W1 contrast on approval pairs is $+0.004$ $[-0.001, 0.008]$. Against EndorseRank with activity restarts the picture is the same (-0.022 $[-0.042, -0.0004]$ and -0.044 $[-0.063, -0.025]$; $+0.164$ $[0.142, 0.187]$ and $+0.164$ $[0.144, 0.186]$). Against AWP the signs reverse: C-PR leads on new approval pairs ($+0.044$ $[0.029, 0.058]$; $+0.040$ $[0.026, 0.055]$, the F1 of S1) and trails on new senders (-0.077 $[-0.087, -0.068]$; -0.077 $[-0.086, -0.069]$). Among the spenders with a payment in-edge at the freeze, 4,569 in W0 and 4,642 in W1, its lead on approval pairs is smaller in W0 ($+0.026$ $[0.013, 0.041]$) and its interval contains zero in W1 ($+0.007$ $[-0.008, 0.021]$), while it trails AWP on new senders by more (-0.084 $[-0.097, -0.069]$; -0.094 $[-0.108, -0.079]$).

The same-window families give the matching picture. C-PR agrees with the transfer family far better than EndorseRank does ($+0.441$ $[0.423, 0.457]$) and with the authorization family better than AWP does ($+0.206$ $[0.188, 0.224]$), and it agrees worse than each on that walk's own family (-0.406 $[-0.419, -0.393]$ against EndorseRank; -0.017 $[-0.028, -0.006]$ against AWP). C-PR thus sits between the two single-layer walks on every comparison: it gains part of each walk's strength on the other walk's label and gives up part of each walk's strength on its own.

VII. SYBIL EXPOSURE OF THE AUTHORIZATION LAYER

The results above concern honest data. A ranking must also be judged by what it costs to fake. This section calibrates the farm results of Gyöngyi and Garcia-Molina [32] to EndorseRank's authorization walk with uniform restarts, on its graph of contracts at the freeze of 30 June 2026 (109,381 nodes and 114,997 edges). The closed-farm bound (9) holds for any uniformly restarted walk on this layer, C-PR at $\lambda = 1$ included, but the star counts below are read against the cohort's EndorseRank scores. The model was computed after the W1 labels were known and decides no rule.

A. COST OF A FARM CONTRACT

Every node is a contract, so a farm address cannot be a free key pair: it must be deployed. In the fee schedule a creation costs 32,000 gas or more if a factory contract performs it and 53,000 gas or more if a transaction of its own does, and every byte of code adds 200 gas [51]. These figures are execution gas, without the rollup's fee for posting data to Ethereum; we did not measure fees in currency. The approval of a farm contract counts only up to what that contract holds of the token at the freeze. A farm contract that approves therefore needs a balance, and an approval from an empty contract is no edge. The balance need not be large. The bounded weight reaches $V = 1.000$ at one dollar, and after row normalization a single out-edge carries the whole row whatever its amount. The balance is not spent: it has to be there at the freeze and can be withdrawn afterwards. Because the graph keeps the approvals set in the twelve months before each freeze, a farm re-emits its approvals at least once a year; inside a closed farm the age discount of EndorseRank does not matter, because each row of a farm contract is normalized over edges that all stay in the farm. Over q such periods a closed farm $\mathcal{S}$ costs

$$C = |\mathcal{S}|\, c_{\text{dep}} + q\left(|E_{\mathcal{S}}|\, c_{\text{app}} + c_{\text{rev}}\right), \tag{8}$$

where c_{dep} is the fee of one deployment, c_{app} the fee of one approval and c_{rev} a per-period allowance for revocations or detection avoidance, beside a balance of about a dollar per approving contract that the farm keeps. Whatever the fees, the cost is linear in contracts, edges and periods. A farm of contracts is thus dearer than a farm of free addresses by its deployments and by the dollars it leaves idle.

B. WHAT A CLOSED FARM COLLECTS

Let every node of a farm $\mathcal{S}$ have out-edges, all inside $\mathcal{S}$, and let no honest owner approve a farm contract. Write r_D for the stationary mass on dangling nodes, all of which are honest. Summing (3) over $\mathcal{S}$ with uniform $\boldsymbol{\pi}$ gives

$$\sum_{i\in\mathcal{S}} r_i = \frac{|\mathcal{S}|}{|V|}\cdot\frac{1-d+d\,r_D}{1-d} \le \frac{|\mathcal{S}|}{(1-d)|V|+d|\mathcal{S}|}, \tag{9}$$

where the bound uses $r_D \le 1-\sum_{i\in\mathcal{S}} r_i$. The second factor is the multiple of its population share that the farm captures, and it exceeds one whenever dangling nodes exist, because their redistributed mass reaches the farm. On the contract graph the dangling mass is large, since most nodes are spender contracts that approve no one: $r_D = 0.911$ at $d = 0.85$ and stays at 0.911 and 0.910 at $d = 0.75$ and 0.95. A closed farm therefore captures 3.73, 6.16 and 18.30 times its population share at the three damping values, and a ring of 20 contracts, each approving the next, inserted into the graph captures 6.16 times its share at $d = 0.85$. On the registered graph, whose nodes are all accounts of the cohort's events, the factors are 2.01, 2.96 and 6.62.

C. CONCENTRATING THE MASS

The bound limits a farm's total mass, but the attacker can concentrate it. In a star, f farm contracts each approve a target t and t approves each of them back. With $c = (1-d+d\,r_D)/|V|$, the mass each node receives from jumps and dangling rows, the balance equations give

$$r_t = \frac{c\,(1+d\,f)}{1-d^2}. \tag{10}$$

TABLE 9. C-PR Beside Every Other Score and Count of the Main Analysis: $'\tau = \tau_{\text{C-PR}} - \tau_b$ With 95% Paired-Bootstrap Intervals

	W0 spenders ($n = 7{,}864$)		W1 spenders ($n = 7{,}915$)	
C-PR ($\lambda = 0.5$) minus	New approval pairs	New transfer senders	New approval pairs	New transfer senders
Authorization walk, C-PR ($\lambda = 1$)	-0.025^a [−0.045, −0.004]	+0.170 [0.148, 0.191]	-0.043^b [−0.061, −0.024]	+0.173 [0.152, 0.194]
Payment walk, C-PR ($\lambda = 0$)	+0.071 [0.059, 0.084]	-0.014^a [−0.019, −0.008]	$+0.074^b$ [0.060, 0.087]	-0.016^b [−0.022, −0.012]
EndorseRank	−0.026 [−0.046, −0.004]	+0.170 [0.148, 0.191]	−0.043 [−0.062, −0.025]	+0.172 [0.152, 0.193]
EndorseRank, activity restarts	−0.022 [−0.042, −0.0004]	+0.164 [0.142, 0.187]	−0.044 [−0.063, −0.025]	+0.164 [0.144, 0.186]
AWP	+0.044 [0.029, 0.058]	−0.077 [−0.087, −0.068]	$+0.040^s$ [0.026, 0.055]	-0.077^s [−0.086, −0.069]
S-PR	+0.051 [0.038, 0.064]	−0.011 [−0.021, −0.0004]	+0.049 [0.036, 0.063]	−0.006 [−0.015, 0.002]
In-approve degree	−0.062 [−0.082, −0.037]	+0.163 [0.142, 0.187]	−0.077 [−0.097, −0.057]	+0.164 [0.141, 0.188]
In-approve value	−0.048 [−0.068, −0.024]	+0.173 [0.152, 0.195]	−0.059 [−0.079, −0.039]	+0.174 [0.151, 0.197]
Transfer in-degree	+0.072 [0.059, 0.086]	−0.064 [−0.075, −0.054]	+0.072 [0.058, 0.087]	−0.066 [−0.075, −0.057]
Union in-degree	+0.016 [0.007, 0.024]	−0.053 [−0.061, −0.043]	+0.011 [0.003, 0.019]	−0.056 [−0.065, −0.047]
Spenders that another contract had paid before the freeze (W0 $n = 4{,}569$, W1 $n = 4{,}642$)				
AWP	+0.026 [0.013, 0.041]	−0.084 [−0.097, −0.069]	+0.007 [−0.008, 0.021]	−0.094 [−0.108, −0.079]
In-approve degree	+0.051 [0.025, 0.076]	—	+0.053 [0.025, 0.080]	—

A negative value means that C-PR ranks the label below the other score. The counts are taken on the contract graph at the freeze. [a]The two contrasts of rule A. [b]F6a, F1 and F2 of the registration of W1. [s]F1 and F2 of its first sensitivity analysis. These rules were fixed for the registered analysis and are applied here to the scores of the contract graph, as a re-analysis. The contrasts with the union in-degree belong to the definition of the main analysis and to no rule; every other cell was computed after the labels were known and belongs to no rule. No cell is a registered result. The lower block repeats two rows on the spenders that AWP can order, a restriction also chosen after the results were known. Within a cohort every cell uses the same 400 resamples (seed 42).

On the contract graph $c = 8.45 \times 10^{-6}$, the common score of every node without in-edges. With $f = 10$, eleven contracts and twenty approvals per period, the target scores 2.89×10^{-4} on the graph with the star added, the value of (10) and 3.0% of the highest EndorseRank score in the cohort (0.0097). Each further farm contract adds $d\,c/(1 - d^2)$ at the price of one deployment and two approvals per period. Solving (10) for f shows what each rank costs. The low ranks cost almost nothing, because only 1,243 of the 6,443 cohort contracts have a positive EndorseRank: the cohort's median score is zero and its 90th percentile (9.87×10^{-6}) lies just above c, so a pair of contracts that approve each other, which scores $c/(1-d)$, passes both. The 99th percentile (9.31×10^{-5}) takes 3 farm contracts, and the highest score among the cohort's contracts about 374, a count that ignores how a farm that large would itself enlarge $|V|$ and lower c. On the registered graph the 90th and 99th percentiles and the top took 13, 1,253 and about 1,080,271 farm addresses, none of which needed a deployment or a balance. The contract graph makes each farm address dearer, but it is small, and its top is within reach of a few hundred contracts.

D. A BALANCE THRESHOLD

The balance cap suggests a defence: count an allowance only when its capped value reaches a threshold θ, so that every farm contract that approves must hold at least θ. Table 10 shows what this does on the observed graph. A threshold of one dollar keeps 3.7% of the pairs (4,247 of 114,997), and the cohort's ranking then agrees with the main one at $\tau_b = 0.709$; ten dollars keep 1.9% and a hundred dollars 0.9%. The farm is hurt less than the honest graph. The star of 10 must hold 11, 110 or 1,100 dollars in all, money it keeps, while its target rises from 3.0% of the cohort's highest score to 9.2%, 32.5% and 105.4%, and the farm that reaches the top shrinks from 374 contracts to 119, 33 and 9. The reason is that a farm of a given size is a larger part of a smaller graph: c grows from 8.45×10^{-6} to 7.74×10^{-4} as the nodes fall from 109,381 to 837. At the sizes examined a balance threshold therefore raises the capital a farm needs and lowers the number of contracts it needs, and it buys no protection; larger thresholds were not examined.

TABLE 10. Farms of Contracts on the Authorization Graph at the Freeze of 30 June 2026, Without and With a Balance Threshold θ

Threshold θ	None	$1	$10	$100
Authorization pairs kept	114,997	4,247	2,167	992
share of the pairs	100%	3.7%	1.9%	0.9%
Owner contracts	9,452	2,390	1,000	500
Nodes of the graph	109,381	3,394	1,769	837
Cohort contracts with a positive score	1,243	699	491	308
τ_b of the cohort's scores with the main ones	1	0.709	0.589	0.464
Dangling mass r_D	0.911	0.517	0.582	0.586
Target of a star of 10: share of the cohort's highest score	0.030	0.092	0.325	1.054
Farm contracts that reach the cohort's highest score	374	119	33	9
Balance the 11 contracts of the star hold (USD)	—	11	110	1,100

EndorseRank graph of contracts in the main analysis. With a threshold the graph keeps only the pairs whose capped value is at least θ, so every farm contract that approves must hold θ of a token at the freeze. A farm address is a contract: its creation costs at least 32,000 gas from a factory and 53,000 gas by a transaction of its own, plus 200 gas per byte of code (execution gas, without the fee for posting data to Ethereum). The farm sizes come from the closed form (10) and count the farm contracts beside the target. Computed after the W1 labels were known; the model decides no rule.

E. THE SAME ARGUMENT FOR THE COUPLED WALK

In C-PR at any $\lambda > 0$, the renormalization in (5) lets a node whose out-edges are all approvals follow them with weight one. A farm whose contracts approve only one another and

make no payments is therefore closed in C-PR exactly as in the authorization walk, and (9) applies with C-PR's dangling mass. We did not compute that mass or run the farms on C-PR's graph, so the multipliers above are not claimed for it; the argument shows only that coupling does not close the gap. An approval bought from an honest owner contract shifts mass in proportion to its share of that owner's row. In EndorseRank's bounded layer a fresh approval at full value weight takes 68.7% of the median owner contract's row.

F. REMEDY, UNTESTED

The asymmetry that Cheng and Friedman identify points to the fix [33]. When jumps and the mass of dangling nodes are confined to a vetted seed set, as in TrustRank [13], a farm outside the seeds gains mass only along edges from honest owners, and those edges must be bought. We did not evaluate seeded restarts, and S-PR is not such a remedy, because its seeds are authorization scores that a farm can inflate. Credential gates such as Human Passport [23] address a different question, whether an address belongs to a distinct person, which has no direct counterpart for a contract, and they carry their own weaknesses [39]. The analysis is model-based and was checked with the solver on the observed graph; no attack was run on chain.

VIII. THREATS TO VALIDITY

A. CONSTRUCT VALIDITY

The labels are arrivals of the same kinds of edge that build the layers, so they measure future attachment and favor scores that track current counts; neither is a safety or loss outcome, and "the contracts users rely on" is measured only through them. They count mostly user accounts, which no score reads, so a score is judged on demand it cannot see. The node rule depends on detecting smart accounts. It finds Safes and ERC-4337 accounts. With the listed code families applied, 115,084 addresses typed as contracts still have code with a `validateUserOp` selector but without a paymaster selector, and contracts that a service creates per user, payment or order are not singled out either. Where such contracts are nodes, the graph holds accounts of the kind that the node rule is meant to keep out. Types are read from code at one block after the end of the data. The authorization edge is the latest approval in the score window, capped at the owner's balance at the freeze, not the allowance in force (Section III); a balance held only at the freeze counts in full.

The authorization layer between contracts is thin. It has 115,006 pairs against 2,580,709 payment pairs at the freeze of 30 June 2026 and 17,940 against 2,269,073 at that of 31 March 2026, and a fifth to a quarter of each cohort is an end of an authorization pair, so every result on the authorization side rests on few contracts. At the freeze of 30 June 2026 one owner contract holds 98,073 of the 115,006 pairs. The W1 scores, the same-window checks and the Sybil calibration are computed on a layer that this one contract largely shapes; we neither identified it nor repeated the analysis without it, and the two freezes differ so much in this layer that W0 and W1 are not replications of each other on the authorization side. Coverage limits the other side: the contracts outside the union graph, about 27% of each spender cohort, are tied at zero by construction, and Table 8 shows that two contrasts read differently without them. The two analyses differ in graph, weighting, tokens and score windows at once and are not a sweep over any of them, and other couplings, such as layers with EndorseRank's and AWP's bounded weights, are not evaluated in this paper.

B. INTERNAL VALIDITY

Rule A was fixed before any log of this study was extracted. Rule B's hypothesis came from an exploratory W0 contrast, and its margin from a W0 interval. The registration is a commit in the authors' own repository, not an entry in an independent registry, and its precedence over the W1 extraction rests on commit and job times rather than on a third-party timestamp (Section IV-C). The main analysis was defined after every registered result was known and with the labels of both windows known, so its choices could favor one outcome. Two facts limit that risk without removing it: its constants come from quantiles of events in the twelve months up to the W0 freeze, so no event of a label window enters them, and it gives the same readings of both rules as the registered analysis, neither met, with F1 holding and F2 failing. Reporting it first is likewise a choice made after the results; the registered analysis stays in the tables and figures of the rules as the test that the rules were fixed for. Rule B is read here through two of its criteria, which can show that it is not met and could not have shown that it is. The rule that scores absent contracts as zero is a pipeline choice; S2 shows that it changes no reading, and Table 8 shows what it does to the contrasts with the authorization walk.

C. STATISTICAL CONCLUSION VALIDITY

Rule-bound intervals come from 400 paired resamples, as fixed in the plan. Across 50 further seeds no verdict of A1, A2, F1 or F2 changes in either analysis; the interval end nearest to its threshold, the upper end of A1 in the main analysis, stays below zero in every seed (Appendix A). The many contrasts reported beside the rules are nominal and should not be read as a family of tests. The contract bootstrap treats spenders as independent although one new owner can approve several spenders. The labels are sparse: 1,491 of the 7,915 W1 spenders gained a new approval pair and 1,369 a new sender, so each coefficient rests largely on how those subsets are ordered.

D. EXTERNAL VALIDITY

The evidence comes from one rollup, five tokens, contracts approved by at least three owners, scores built from the year before each freeze (from every event since October 2023 in the registered analysis), two three-month label windows and account types read at one block. A graph of contracts on another chain, or with other tokens, can have a thicker authorization layer and behave differently. Only fixed global weights

with uniform teleportation were tested; node-dependent or learned weights, mixtures of stationary vectors, interlayer copies of nodes and per-layer damping remain open.

IX. IMPLICATIONS FOR WALLETS, EXPLORERS AND INTEGRATIONS

A wallet that warns before an approval, an explorer that orders contracts and a team that chooses contracts to integrate can take five points from these results, each at the size the results support. First, the coefficients are small. The best score on each label reaches about 0.2 on later approvals and about 0.3 on later payers, on labels that count mostly user accounts, so any of these scores is a weak hint for a human reader and not a basis for an automatic decision. Second, the cheapest scores were hard to beat. Counting the owner contracts that hold a capped allowance to a contract anticipated its later approvals better than every walk in this study, the coupled walk included, and counting the contracts that pay it anticipated its later payers better than the coupled walk, with no solver and no restart vector to defend. AWP was slightly ahead of the payer count on the whole cohorts and, in the first window, behind it among the contracts it can order. Third, folding the two relations into one stationary distribution did not help: mixing an authorization layer into a payment walk moved the ranking mainly by switching the layer on, and the gain on later approvals came with a loss on later payers larger than the registered margin. An interface that wants both signals can show the two counts, or AWP and an approval count, side by side. Fourth, a ranking read from edges between contracts is silent on the contracts that only user accounts touch. About 27% of each spender cohort has no such edge, and three quarters have no authorization edge; an interface should show these contracts as unranked, not as ranked low. Fifth, a score that walks authorization edges with uniform restarts can be inflated by a closed farm of contracts. Requiring a deployment and a balance for every farm address raises the cost per address, but a dollar of balance suffices, a few hundred farm contracts reach the top of the cohort, and a balance threshold of up to a hundred dollars does not protect the layer (Section VII). Restricting restarts to vetted seeds is the remedy the theory points to, but we did not test it. Beside these points, approval logs are not allowance state: most positive allowances between contracts are backed by no balance at the freeze, and an integration that acts on an allowance should read the allowance and the balance from contract state at the decision block.

X. CONCLUSION

Authorization and payment give two readings of the same contracts. On graphs whose nodes are contracts only, a walk that blends the two at every node did not beat either layer walked alone. Under the rule fixed in the plan before this study extracted any log, applied to the contract graph as a re-analysis, C-PR ranked later approvals below the authorization walk (-0.025 $[-0.045, -0.004]$) and later payers below the payment walk (-0.014 $[-0.019, -0.008]$) in the first window. In the window registered before this study extracted it, C-PR repeated its gain over the payment walk on later approvals ($+0.074$ $[0.060, 0.087]$) but ranked later payers worse by more than the registered margin (-0.016 $[-0.022, -0.012]$), and against AWP by more still; Rule B is not met. The registered analysis in raw token units, the test that both rules were written for, met neither rule either ($+0.101$ $[0.092, 0.111]$ and -0.025 $[-0.029, -0.021]$ on the two criteria). The gain on approvals appears as soon as the authorization layer is switched on and changes little across interior weights, and seeding the payment walk with authorization scores recovered only part of it. In contrasts made afterwards C-PR lay between the single-layer walks, above AWP and below EndorseRank on later approvals and the reverse on later payers. The counts were hard to beat: C-PR ranked later approvals below the count of approving contracts and later payers below the count of paying contracts in both windows, and against the count of both kinds of edge it led slightly on approvals and trailed on payers. These numbers describe contracts that other contracts pay or approve; about 27% of each spender cohort has no such edge, and the labels count mostly user accounts. An authorization layer of contracts costs more to farm than one of free addresses, since each farm address needs a deployment and a balance, but a dollar of balance suffices, a few hundred farm contracts reach the cohort's top score, and coupling does not remove that exposure. Future work should fix in advance, anchor with a third-party timestamp and test on unextracted data couplings with comparable weights on both layers, importance labels from outside the chain, a better detection of smart accounts, and seed-restricted restarts under explicit attack. Whether such a ranking could later serve as one input to credit decisions about contracts was not tested here.

APPENDIX A EXPLORATORY CHECKS

All checks below were run after every result of both analyses was known, by the script in the supplement, which reads the published per-contract score files and summaries and builds no graph. None of them enters a rule, and none is a registered result. Table 11 lists the results.

Recomputation. From the score files the script recomputes A1, A2, F1, F2, F6 and S1 in both analyses and, in the main analysis, the contrasts of C-PR with the union in-degree and the contrasts on the spenders inside the union graph, with the pipeline's resampling and SciPy's τ_b; every recomputed value equals the published one. S2 cannot be recomputed, because its isolated-node scores are not in the score files.

Seed spread. Across 50 further seeds of 400 resamples, the verdicts of A1, A2, F1 and F2 do not change in either analysis (Table 11(a)). The interval ends have standard deviations of 0.0002 to 0.0012. The upper end of A1 in the main analysis, which decides "no worse", ranges from -0.0078 to -0.0031 and stays below zero, and the upper end of F2 ranges from -0.0121 to -0.0102, so that the whole interval lies beyond the margin in every seed. On the spenders inside the union graph the interval of A1 contains zero in 49 of the 50 seeds

TABLE 11. Exploratory Checks, Computed After Every Result of Both Analyses Was Known

		Contract-graph analysis (main)	Registered analysis
(a) Rule contrasts under 50 further seeds of 400 resamples: interval end, mean (SD) [range]; seeds with another verdict than seed 42			
A1, W0, new approval pairs	upper end	−0.0060 (0.0011) [−0.0078, −0.0031]; 0 of 50	0.0132 (0.0004) [0.0124, 0.0142]; 0 of 50
A2, W0, new transfer senders	lower end	−0.0192 (0.0004) [−0.0202, −0.0186]; 0 of 50	−0.0336 (0.0002) [−0.0342, −0.0329]; 0 of 50
F1, W1, new approval pairs	lower end	0.0617 (0.0007) [0.0604, 0.0633]; 0 of 50	0.0926 (0.0006) [0.0905, 0.0939]; 0 of 50
F2, W1, new transfer senders	lower end	−0.0215 (0.0004) [−0.0224, −0.0209]; 0 of 50	−0.0281 (0.0003) [−0.0287, −0.0275]; 0 of 50
	upper end	−0.0110 (0.0004) [−0.0121, −0.0102]; 50 of 50†	−0.0209 (0.0002) [−0.0214, −0.0204]; 50 of 50†
A1 in the union graph, W0	lower end	−0.0030 (0.0017) [−0.0079, 0.0001]; 49 of 50‡	—
F6a in the union graph, W1	lower end	−0.0161 (0.0011) [−0.0184, −0.0130]; 50 of 50‡	—
(b) Steps of the weight: τ_b of C-PR at the higher λ minus at the lower one, paired, 400 resamples, seed 42			
W0, new approval pairs	$0 \to 0.25$	+0.070 [0.058, 0.082]	+0.092 [0.083, 0.101]
	$0.25 \to 0.75$	+0.001 [−0.0001, 0.002]	+0.002 [0.0001, 0.003]
	$0.75 \to 1$	+0.026 [0.004, 0.045]	−0.008 [−0.013, −0.003]
W0, new transfer senders	$0 \to 0.25$	−0.011 [−0.016, −0.006]	−0.024 [−0.027, −0.020]
	$0.25 \to 0.75$	−0.005 [−0.006, −0.004]	−0.013 [−0.015, −0.012]
	$0.75 \to 1$	−0.167 [−0.189, −0.145]	−0.095 [−0.102, −0.087]
W1, new approval pairs	$0 \to 0.25$	+0.073 [0.058, 0.086]	+0.099 [0.090, 0.109]
	$0.25 \to 0.75$	+0.002 [0.0005, 0.003]	+0.002 [0.001, 0.003]
	$0.75 \to 1$	+0.043 [0.024, 0.060]	−0.010 [−0.014, −0.006]
W1, new transfer senders	$0 \to 0.25$	−0.013 [−0.019, −0.009]	−0.020 [−0.023, −0.016]
	$0.25 \to 0.75$	−0.005 [−0.006, −0.004]	−0.012 [−0.013, −0.010]
	$0.75 \to 1$	−0.170 [−0.191, −0.150]	−0.095 [−0.102, −0.088]
(c) Rank agreement: τ_b between the C-PR scores at two values of λ over the cohort, 400 resamples, seed 42			
Matched cohort	$\lambda = 0$ and 0.25	0.896 [0.887, 0.907]	0.624 [0.616, 0.632]
	$\lambda = 0.25$ and 0.75	0.983 [0.981, 0.984]	0.869 [0.867, 0.871]
	$\lambda = 0.75$ and 1	0.459 [0.443, 0.472]	0.618 [0.612, 0.624]
W0 spenders	$\lambda = 0$ and 0.25	0.892 [0.883, 0.901]	0.598 [0.592, 0.605]
	$\lambda = 0.25$ and 0.75	0.979 [0.977, 0.980]	0.884 [0.882, 0.886]
	$\lambda = 0.75$ and 1	0.456 [0.443, 0.467]	0.658 [0.652, 0.663]
W1 spenders	$\lambda = 0$ and 0.25	0.891 [0.881, 0.901]	0.594 [0.586, 0.600]
	$\lambda = 0.25$ and 0.75	0.978 [0.977, 0.979]	0.884 [0.882, 0.886]
	$\lambda = 0.75$ and 1	0.460 [0.449, 0.473]	0.660 [0.655, 0.665]

Computed from the per-contract score files of the pipeline after every result of both analyses was known; no row enters a rule and none is a registered result. The script first recomputes the contrasts of Table 7 except those of S2, the contrasts of C-PR with the union in-degree and those inside the union graph (Table 8), and reproduces the published values exactly. In (a) the A1 row reports the upper end, which decides "no worse". †Seeds in which the whole F2 interval lies beyond the margin -0.01. ‡Seeds in which the interval contains zero, on the spenders inside the union graph.

and that of F6a in all 50, so the reading that C-PR and the authorization walk are not separated there does not rest on the seed.

Steps of the weight. Switching the authorization layer on (λ from 0 to 0.25) raises τ_b on new approval pairs by 0.070 to 0.099 and lowers it on new senders by 0.011 to 0.024, with intervals that exclude zero in both windows and both analyses (Table 11(b)). Between $\lambda = 0.25$ and 0.75 the change is at most 0.002 on new approval pairs and between -0.005 and -0.013 on new senders. The last step, from 0.75 to 1, separates the analyses. In the main analysis it raises τ_b on new approval pairs ($+0.026$ $[0.004, 0.045]$ in W0; $+0.043$ $[0.024, 0.060]$ in W1) and lowers it on new senders by 0.167 and 0.170; in the registered analysis it lowers both, by 0.008 and 0.010 on new approval pairs and by 0.095 on new senders.

Rank agreement. In the main analysis the C-PR rankings at $\lambda = 0.25$ and $\lambda = 0.75$ agree at τ_b between 0.978 and 0.983 on every cohort, and the payment walk agrees with $\lambda = 0.25$ at 0.891 to 0.896, whereas the authorization walk agrees with $\lambda = 0.75$ only at 0.456 to 0.460 (Table 11(c)). With so few nodes in B, the interior of the family is close to the payment walk and far from the authorization walk. In the registered analysis the interior values agree at 0.869 to 0.884, and each end point agrees with its nearest interior value at 0.594 to 0.660.

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AUTHOR CONTRIBUTIONS

Taehong Kwon: conceptualization, methodology, software, data curation, formal analysis, writing – original draft. Ernest Mnkandla: supervision, writing – review & editing. Donatien Koulla Moulla: supervision, writing – review & editing. David Sena Attipoe: supervision, writing – review & editing.

CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AVAILABILITY

The study uses only public blockchain data, public daily token prices and the public Chainlink feed directory, and involves no human participants. The code, configuration, plan, registration, the definitions of the main analysis (`docs/revision_contract_graph.md`), extraction manifest, cost ledger and machine-readable result summaries with their SHA-256 checksums are in the folder `experiments` of the public repository https://github.com/abitocodes/ERC20-Allowance-PageRank-Wallet-Reputation, whose commit history is described in Section IV-C. The registered results are in `data/3-published-results-for-thesis/` and those of the main analysis in its subfolder `revised-contract-12m/`, committed in `6b389df` and computed at code revision `bb78da3`. The per-contract score files and the tables of the contract graph are in a public Google Drive folder whose address and checksums are listed in `experiments/drive-manifest.json` at commit `7aa7928`. The extracted event tables are kept in the authors' BigQuery project and can be rebuilt from the public log table with the committed queries. This manuscript's source, its table and figure scripts and its supplement (five queries of the registered analysis, a list that locates the queries of the main analysis in the public repository at commit `6b389df`, and the exploratory-check script with its output) are in https://github.com/abitocodes/phd-dissertation-draft-en, folder `manuscripts/B-ieee-access`. The scores rank contracts only; user accounts are nodes of no graph and enter the labels only as counts, no off-chain identity was collected or linked, and the paper reports aggregate results only.

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